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Design Investigation of a Marine Current Turbine Having a Winglet on the Blade

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To God be the glory!

ABSTRACT

The energy profile of current speeds in the bulk of Southeast Asia including the Philippines is lower than its western counterparts where most commercial tidal turbine blades are designed. Thus, a different way of improving and optimizing the efficiency of said turbines could be done such as improving the rotor blades geometry or by the application of winglets to increase the power coefficient and reduce turbine tip vortices. Two winglet designs, trapezoidal and triangular shaped, were applied towards a base rotor blade. After confirming that the hydrodynamic model agrees well with the experimental tests, numerical simulations were done to test the effectiveness of such winglets. Both winglet designs demonstrate an increase in power coefficient with an average increase of 3.71% for the trapezoidal configuration and 2.86% for the triangular configuration and a higher increase of power coefficient at higher TSR. The increase in performance can be attributed to an increase in pressure difference between the suction and pressure side of the blade near the blade tip. The winglets were also able to reduce tip vorticity.

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I. INTRODUCTION

1.1 Background

The energy supply of the Philippines is still dominated by fossil fuels. According to the Philippine Energy Plan 2020 - 2040 of the Department of Energy, as of 2020 the share of renewable energy is sitting at 21.24% despite the goals set in the past years. Hydropower contributes 7.07% of the total energy share while wind energy contributes 1.01%. Shown on Table 1 is the fuel distribution for 2020-2040.

Table 1.1 Gross Generation Output by Fuel (TWh) from Philippine Energy Plan 2020-2040

Fuel Type	2020		2030		2040		AAGR(%)
	Levels	%Shares	Levels	%Shares	Levels	%Shares	'20-'40
Coal	58.18	57.17	87.75	15.16	89.72	24.62	2.19
Natural Gas	19.5	19.16	37.51	19.30	146.86	40.30	10.62
Oil	2.47	2.43	1.03	0.53	0.28	0.08	-10.39
Renewable	19.29	21.24	68.04	35.01	127.54	35.00	9.28
Geothermal	10.76	10.57	15.63	8.04	16.18	4.44	2.06
Hydro	7.19	7.07	15.87	8.17	51.55	14.15	10.35
Wind	1.03	1.01	3.31	1.70	5.12	1.41	8.37
Solar	1.37	1.35	32.06	16.50	53.06	14.56	20.05
Biomass	1.26	1.24	1.16	0.60	1.63	0.45	1.29
Total	101.76	100.00	194.33	100.00	364.40	100.00	6.59

Currently, the country's main source of energy is from non-renewable sources, wherein more than 75% of the gross generation output of fuel comes from, and this is why the Philippine Energy Plan focuses more on reducing the share of these sources and improving more on the renewable sources of energy. One of the popular renewable energies is wind power. As suggested by its name, it utilizes the wind, particularly the wind speed available in an area to harness wind

energy. Even though it is lower in shares on the country's gross generation output, it is the popular option amongst people as it makes the landscape more visually appealing since they are large objects (US EIA, 2021). Additionally, they also do not have toxic emissions and do not generate waste during operation of the turbines (Acciona, 2017). Operationally, wind turbines work by converting the kinetic energy of the wind to mechanical energy upon flowing onto the rotor blades and is essentially converted into electricity via a generator to be used for other purposes (Danao et al., 2020). Similarly, this is how tidal turbines work with the exception that instead of wind, water is in consideration.

Furthermore, among the available renewable energy sources in the country, Hydropower, particularly the tidal generation of energy, is the one of the most advantageous as it is predictable in long periods of time, unlike wind behavior that is relatively unpredictable thus reducing its efficiency and potential energy output. It is also second on the highest renewable energy source of the country. Moreover, its operation is cost-efficient as it requires no fuel to work and is not harmful in terms of its emissions (Denny, 2009). In current times, there are two methods to harness tidal energy: tidal range devices and tidal stream devices. The former generates electricity by taking advantage of the water level difference, while the latter utilizes the flow of water in tidal currents (Roberts et al., 2016). Both of these turbines take advantage of the kinetic energy of the medium, which is similar to how wind turbines work. The only problem with this technology is that it needs to have a specific location with regards to current velocities in order to be commercially viable (Fairley et al., 2014).

Locally, according to Solenergy Systems Inc., the Philippines has two methods of harnessing hydropower, one through dams, and the other is through run-of-river, which basically considers flowing bodies of water. There are multiple hydroelectric power plants found in the country, but for tidal stream devices, a particular location has been in consideration due to having high flow speeds. The San Bernardino Strait, a narrow passage on the north-eastern side of the Philippine archipelago, separating the main islands of Luzon and Samar (Battaglia, 2018), has been identified as a potential site for tidal stream energy converters with peak current speeds reaching 4.5 m/s (Jones et al., 2011), however the average annual current is only between 1.20 -

1.60 m/s (Ibanez et al., 2018). Other sites in the Philippines have even less current speeds with magnitudes between 0.40 - 0.80 m/s (Bausas et al., 2016).

Moreover, other than changing the speed of either the wind or current, another way to improve and optimize the efficiency of turbines is by improving its rotor blades' geometry. Numerous studies have been done such as having composite makeup along the blade where thicker blades are put nearer the turbine hub or optimizing the angle of attack or pitch along the blade. Also, historically, the capacity factor is a consideration for the turbines, and has been steadily increasing as time progresses since the production of bigger blades are more possible. One of further developments are winglet blades for such turbines, since they are commonly known for aircraft use due to having an increase in lift from the reduction of induced drag. The addition of winglets as demonstrated by Maughmer (2003) in sailplanes, provides a gain in overall performance and gives insights, but requires further investigation, in applications to tidal turbines.

1.2 Problem Statement

The Philippine Energy plan shows an increase in demand for hydropower in subsequent decades. With regards to tidal stream devices, most commercial tidal turbine blades are rated for speeds above 2 m/s (Roberts et al., 2016). With this, the performance and efficiency of hydropower devices, particularly tidal stream devices, should be considered especially that in the bulk of Southeast Asia including the Philippines, low velocity profiles are found (Quirapas et al., 2015).

1.3 Objectives of the Study

This study aims to numerically investigate the effects of a winglet on the blade of a marine current turbine at different tip speed ratios (TSRs) and compare turbine characteristics, specifically power coefficient and the turbine tip vortices. The study also aims to identify the better winglet configuration between a trapezoidal tip and a triangular tip for a marine current turbine wherein its blades are based on the blade utilized in the study of Bahaj et al. (2007).

1.4 Significance of the Study

Studies that include winglets to increase efficiency of turbines have been more prominent in recent years to increase production of power, but these are more focused for horizontal axis wind turbines. Its hydropower counterpart, horizontal axis tidal stream turbines, have significantly fewer studies that are focused more on the appropriate airfoil for the blade. Since there are fewer discourses regarding the application of winglets on horizontal stream tidal turbines and its effectiveness on such devices, this paper will focus on the study thereof. This study is also in accordance with the Philippine Energy Plan that prioritizes hydropower more than wind energy for the transition of the country to more renewable energy sources.

1.5 Limitations

1. The results of this study only reflect the winglet effect on the blade from Bahaj et al. (2007) only.
2. The flow was assumed to be uniform and steady and is based on the average flow speed in the Philippines.
3. ANSYS Fluent was used for the numerical analysis (CFD) throughout the study and had a limited computing power.
4. The current is assumed to be uniformly flowing axially through the turbine blades.
5. Only the power coefficient, pressure, and tip vortices would be investigated.

II. REVIEW OF RELATED LITERATURE

2.1 Wind turbine studies

Wind energy has been one of the most lucrative and fastest growing energy resources. In the past 30 years, wind energy has grown significantly reaching a global capacity of 651 GW by the year of 2019 (Global wind report 2019, 2021). Operationally, wind turbines work by converting the wind energy to mechanical energy when fluid (wind) flows through the rotor blades. This energy is then converted into electricity via a generator to be used for other purposes (Sy et al., 2020).

The recent advancements on this technology are heavily favored in countries which experience extreme wind conditions such as European and North American countries. These improvements follow international guidelines and standards, but the demands are increasing for countries that are affected by tropical cyclones such as the Philippines (Freudenreich et al., 2014). In 2020, wind energy contributes 1.01% to the country's gross generation output, and according to the report by the International Renewable Energy Agency (IRENA) in 2021, wind energy capacity has remained unchanged since 2019 at 443 MW. But even so, the demand on this energy increases further as the country transitions to more renewable energy options as seen on the Philippine Energy Plan for 2020-2040.

In lieu of this growing trend and demand, increasing the efficiency of the said wind turbines has been an utmost priority for researchers and engineers alike. Optimizing the blade profile has been one of the prominent fields of research on it and according to Scopus as of November 11, 2021, there are 86,279 documents revolving the research pertaining to wind turbines. Another method used was to have a different approach from the typical horizontal axis wind turbines or HAWTs.

HAWTs have its axis of rotation point parallel to the wind direction, but vertical axis wind turbines (VAWT) have its axis of rotation perpendicular to the ground. This gives an advantage of being omni-directional in receiving winds, with HAWTs needing to have a yawing means. Mainly, the disadvantage of VAWTs is that it has a lower ideal efficiency than HAWTs with the addition of being immune to backtracking (Riegler, 2003).

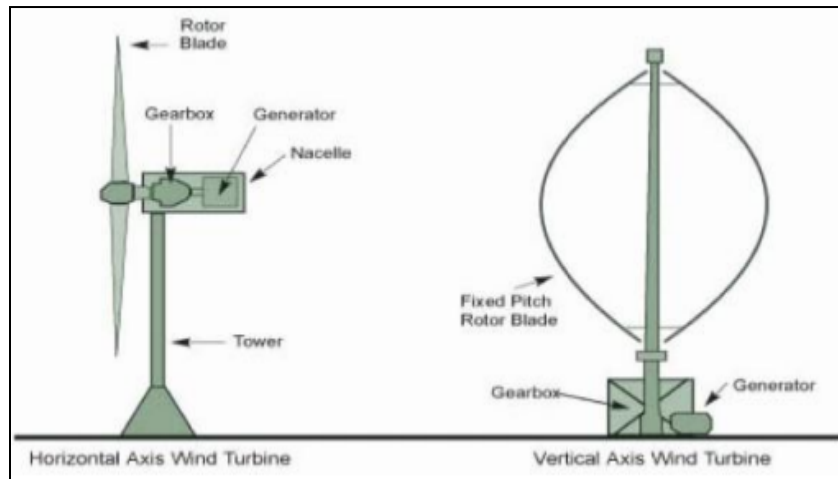


Figure 2.1 HAWT vs VAWT. Image Source: Bidi et al. (2017)

As much as one would like to optimize the harvesting of the said energy, multiple problems arise. Firstly, the country has limited land space due to its nature of being an archipelago, having restricted locations for wind farms. As a solution to this, offshore wind turbines are used wherein they are put above bodies of water. But even so, wind energy itself has unpredictable and intermittent patterns (Tidal Energy, n.d), which is another problem. Furthermore, as clearly shown on the country's Energy Plan, the main objective is to increase the generation output from Solar and Hydropower. Thus, an alternative such as tidal energy for hydropower remains as a lucrative research opportunity due to its advantage of utilizing a much denser energy resource, thus producing more power with the same turbine rotor parameters.

2.2 Tidal turbine studies

Tidal stream turbines (TSTs) provide an alternative way of harnessing energy from the marine currents without needing the enclosure of large areas in maritime space. They are able to provide multiple power generation periods per day, everyday, and with this relatively high frequency power cycling, the said system can partially displace oil generators in islands and lessen our reliability on these systems. (Coles et al., 2021).

Similar to wind turbines, tidal stream turbines have two general types: Horizontal Axis tidal stream turbines (HATST) and Vertical Axis tidal stream turbines (VATST). HATSTs solves

problems encountered by VATSTs such as low starting torque (Khan, 2009) which means that these types of turbines have problems on self-starting. Moreover, contrary to HAWTs, Horizontal Axis Tidal Stream Turbines have generally better efficiency (Satrio et al., 2016) which is why studies as well as commercially available products of HATSTs are more abundant than its vertical axis counterpart. In addition, technological advancements for wind turbines are also broadly applicable for hydro turbines too (Frankael, 2010).

Current commercial turbines such as the one used by SIMEC Atlantis Energy deployed in Scotland employ several 1.5 MW 3-bladed turbines known as the AR1500 (*Tidal turbines: Installation & Procurement: Atlantis resources*, 2019). These commercial tidal turbine blades are proven to have an appropriate thickness near the root of the blade and are able to resist high flap and edgewise bending moment which is needed for the said settings (Gonabadi et al., 2021). However, these commercial blades are rated for speeds two m/s and above and thus a different blade geometry should be determined (Roberts et al., 2016) especially that for the bulk of Southeast Asia (Quirapas et al., 2015), low velocity dominates the region. In addition to this, there is limited literature for low-velocity current turbines since, as previously mentioned, most commercially manufactured or produced tidal turbine blades are for energetic tidal profile.

According to Roberts et al. (2016) the commercially developed tested turbines achieve similar levels of efficiency to those tested in the study even if a huge difference in scale exists. Thus, in order to develop an efficient tidal turbine to be commercially available a numerical study should be initially conducted. Two common methods used for achieving the said studies is through the blade element momentum theory (BEMT) and computational fluid dynamics (CFD). In the study of Batten (2006), BEMT was experimentally validated and the experiments show similar acceptable results to BEMT's findings. But even so, according to McCombes et al. (2009) and Lee et al. (2012), the method has the ramifications that one must assume a uniform inflow over the elements in addition to the fact that there must be zero flow normal to the streamtube boundaries and thus in any event, they recommend a 3D full-scale analysis, such as CFD, in order to receive a more precise power performance prediction and detailed flow analysis of a turbine.

Upon achieving desired solutions on using such methods, an experimental approach should be done as this would show actual operating results of blades and turbines. A study by Bahaj et al. (2007) manufactured blades, as shown on Figure 2, for a tidal turbine using T6082-T6 aluminium alloy with the profile shape of NACA 63-8xx and is interpolated from NACA 63-812, 63-815, 63-818, 63-821, and 63-824 2D coordinates. Also, the blade's chord, thickness and pitch distribution are specified in Table 2. In addition, the rotor hub allows rotation of the blades for a desired hub pitch angle. This blade was tested in a cavitation tunnel that was 5 meters in length, 2.4 meters wide, 1.2 meters high with max flow speed of 8 m/s.



Figure 2.2 Turbine blade and rotor hub from Bahaj et al. (2007)

Tip speed ratios ranging from 4 to 12 were investigated. Upon experimentation, the maximum power coefficient of 0.46 was reached when the hub pitch was angled at 20° , 0° yaw angle, with flow speed of 1.73 m/s. Similar results were also reached at 1 m/s and moreover, TSR values of 5-7 exhibit best performance for the turbine. Other cases in the study were five different hub pitch angles from 15° to 30° , two pitch angles of 20° and 25° , and different tunnel speeds from 0.8 m/s to 2 m/s. Some observed trends were that at higher hub pitch angles, more scatter of data was observed, and power coefficient decreased. Also, as the yaw angle increases for any hub pitch angles, power decreases. In addition, specific tip speed ratios were achieved by varying the rotor's rotation speed while maintaining the tunnel flow speed and was calculated using the equation below:

$$TSR = \frac{\Omega R}{U_T}$$

where Ω is the rotation speed, R is the radius from the center of the hub to the tip of the blade, and U_T is the flow speed.

Since the TSTs operate under water, measures to avoid cavitation which has a negative impact on the performance of hydrodynamic mechanical devices is needed. Considering that most of the power was produced near the blade tip where the revolution speed was high and that low velocity turbines are not able to utilize large torque and relies on an increase of the said revolution speed, one should look for ways to reduce the hydrostatic pressure at the blade tip and at the blade tip vortex (Encarnacion et al., 2019; Lee et al, 2012).

Even though no prior numerical approaches were mentioned in their study, the blade from Bahaj et al. (2007) was adopted and used as the base blade in this study since it has the full definition of the blade's geometry and has experimental results which was utilized for comparison and analysis of its performance with and without winglets. In addition, CFD was utilized to replicate the experiment's power output into a simulation software. The same parameters of hub pitch angle, yaw angle, and flow speed were used to better replicate the experimental environmental conditions in the previously mentioned study. The flow speed of 1.73 m/s in the same study is also a good representation of the Philippine's tidal and current profile since it is just above the average annual current speed of 1.6 m/s.

2.3 Winglet effects to aircraft wings

Whitcomb, an engineer at NASA Langley Research Center, designed a winglet for a first-generation jet transport wing due to the increase in aviation fuel prices. Winglets are small surfaces that are attached on the wingtip as an extension that has the goal to reduce the intensity of the tip vortices that induce drag, thereby increasing lift resulting in an increase of performance. In this study, the design Mach number is 0.78 and the transport wing's design lift coefficient is 0.44. Two winglets are mounted, one upper and one lower one. The upper winglet was attached rearwards for the reason of reducing high velocity wind that would hit the upper

surface of the wing and was analyzed. Furthermore, a lower winglet was added in front of the upper winglet's placement on the wing and was also studied. A figure is shown below showing the configuration. The effects of the upper winglet alone show that the lift-drag ratio increased by 9% and the induced drag was reduced by 20% at the design lift coefficient of 0.44. Reduced improvements were seen at higher values of lift coefficient due to boundary-layer separation. Furthermore, adding the lower winglet has little effect when compared to the effects of the upper winglet. (Whitcomb, 1976).

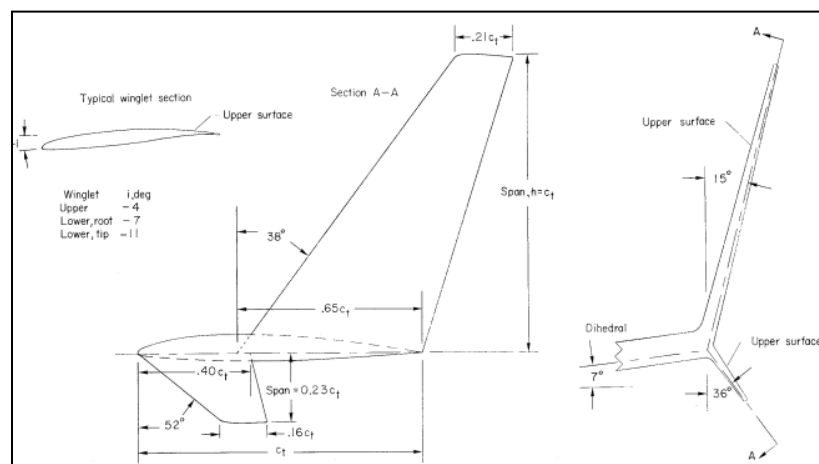


Figure 2.3 Winglet geometry by Whitcomb (1976)

In the study of Maughmer (2003) wherein he designed winglets for high-performance sailplanes have shown that there is an increase in performance wherein induced drag was reduced and even handling qualities was improved. Even though he only achieved a small performance increase, benefits of a properly designed winglet are far reaching, being able to overcome any drawbacks such as reduced properties (e.g. Thermal Strength) or the reduced operating speed of sailplanes. Moreover, an upward-oriented (pointing to suction side) is more effective than using a downward-oriented winglet in reducing induced drag, even though both configurations are beneficial.

As time progressed, different winglets were designed and developed. A study analyzed different winglet designs and their performance, as well as their optimized designs, these were namely: Blended winglet, raked wingtip, split-tip winglet or wing fences, spiroid winglet, and multi-tip winglet (Merryisha and Rajendran, 2019). Their study also had found out that generally

better performances were shown when cambered airfoils were used with winglets with 8° angle of attack, had a sweep angle of between 30° and 50° , twist angle of 0° to 2° , toe angle between -2° and 3° , cant angle of 45° and inclination angle of 5° .

Blended winglets, as shown below, are widely used due to their simplicity. One study shows that using this winglet on a PSU 94-097 airfoil with specifications of the following: a cant angle of 60° and a taper ratio of 0.4, results in increased flight time of 10% due to increase in lift-drag ratio (Panagiotou et al., 2014). Another study using NACA 65218 airfoil had shown that blended winglets at 8° angle of attack cause an increase in L/D ratio by 8.08% (Najafianashrafi and Sedaghat, 2014). Having cant angles of 45° also introduce further improvements in performance, such as a 6% decrease on the drag coefficient on a PSU-90 125WL airfoil at Mach number 0.1465 and Re of 1.5×10^6 (Masud et al., 2016), and good vortex reduction on NACA 65218 under low subsonic flow with an angle of attack of 8° that resulted to a 8% lift coefficient increase (Azlin et al., 2011).



Figure 2.4 Blended winglet on a Boeing 767 Aircraft. Image source: Aviation Partners

Raked winglet/wingtip on the other hand is also considered just a wingtip extension due to the non-existence of a vertical surface. But even so, it is still being used on aircrafts such as the Boeing 767-400 Aircraft (a comparison is shown below). Moreover, the raked wingtip has been proven to increase the lift coefficient and decrease the drag coefficient on angles of attack greater than 5° at different Reynolds numbers by a study made using a tapered NACA 0015

wing, but there was found to be a decrease in effectiveness of the raked wingtip on the drag coefficient as the Reynolds number and angle of attack increases (Samal and Shah, 2014).

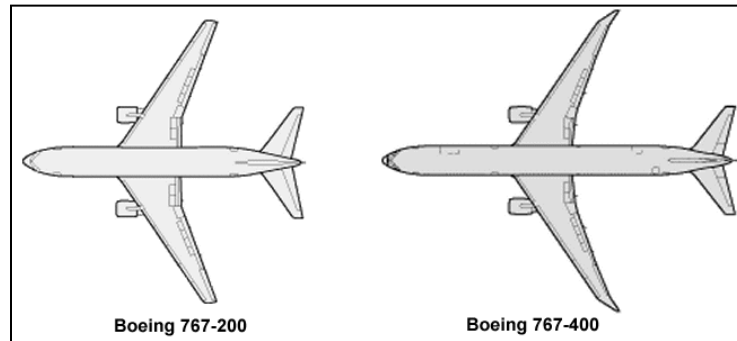


Figure 2.5 Comparison of a Boeing 767 aircraft without and with raked wingtips.

Image Source: AerospaceWeb.org

Another type of wingtip is the split-tip or wing fences. This is similar to Whitcomb's first winglet design but is different since the winglets "split" in terms of their direction, but both connected at the original wing instead of splitting the area on the wingtip, and is shown below. A study on having blended multi-tips has identified that having a split-tip at 65° on the horizontal for both the downward and upward winglet increases the lift coefficient by 10.64% and reduces the drag coefficient by as much as 3.64%. It also reduced the vorticity magnitude at different angles of attack, and in general, reduced aspect ratios increased the aerodynamic performance at 4° angle of attack. Overall, the aerodynamic performance increased by 14% by using blended split-tip winglets. (John and Narayan, 2016)



Figure 2.6 Split-tip winglet on a Boeing 737 MAX. Image Source: Getty Images

Another type is the Spiroid Winglet which has a relatively more complex structure than the other ones as shown below. The problems with these winglets are that it does not eliminate the trailing vortices and drag increases faster at stall. Studies on NACA 0012 show that the induced drag is reduced the best at 8° angle of attack where 75% of this drag is reduced at lift coefficient of 0.95 (Guerrero et al., 2012). Moreover, it also has good performance at 4° on NACA 2412 that results in a depreciation of drag coefficient and 1.5% increase in lift coefficient (Manikandan et al., 2017).



Figure 2.7 Spiroid Winglet on Falcon 50. Image Source: Robin Moret - Daedalum

Finally, multi-tip winglets prevent roll ups to modify the vortex. (Merryisha and Rajendran, 2017). A study that included a split-tip winglet showed that having three tips improved by 22.5% in terms of drag-lift ratio on a NACA 2412 (John and Narayan, 2016). Furthermore, having an angle of attack at 8° also produced optimal results by reducing turbulence intensities (Catalano and Ceron-Muñoz, 2005).

Although this increases the complexity of such objects, the effects of winglets outweigh this as it reduces the cost in terms of aviation fuel for aircrafts in the longer run. Moreover, due to the positive effects of winglets on aircrafts as shown by multiple studies of different winglet designs, researchers extended the application of such onto wind turbine blades. This is because HAWTs in particular use lift force from the wind to turn turbine blades, which is actually similar to how airplanes fly. Since winglets generally increase the lift force for aircraft wings, applying it onto HAWT blades should be investigated because it is also the continuous goal of wind turbine

studies to increase the efficiency of wind turbines, and in particular, optimizing its blades. It should be noted that a suction side-oriented winglet is more effective than its counterpart.

2.4 Winglet application to wind turbines

The similarity of airplane blades to wind turbine blades have brought upon studies to the power-generating device. Both blades experience a similar phenomenon at the blade tip called tip vortex (Khalafallah et al., 2019). This is a three-dimensional effect that occurs due to the difference in pressure on both sides of the blade and leads to reduced lift and generates induced drag, especially on the tip (Gertz et al., 2012). Whitcomb (1976) suggested adding vertical surfaces at the wing tip, wherein it must produce side forces to oppose lift reductions and additional drag, and by using an airfoil cross section on the wingtip produced a vertical diffusion on the tip vortex.

A lot of studies have been conducted with results differing from one another with each of them utilizing different kinds of blades. Elfarra et al. (2015) conducted a numerical analysis of flow properties using an upwind winglet, simulating design parameters of adding sweeping angles, having the cant angles 45° and 90° , twist angles of 0° and 2° , and using wind speeds from 5 m/s to 25 m/s. These design considerations are the factors that Elfarra, Uzol, and Akmandor mentioned that help increase power production in HAWT rotor blades. Results in this study show that tilting the blade tips towards the suction side gave the best performance (positive cant angle) which is similar to what Maughmer et al. (2003) and Whitcomb (1976) suggested. A maximum power increase of 10.43% was achieved at the wind speed of 13 m/s and having a cant angle of 45° and twist angle of 2° .

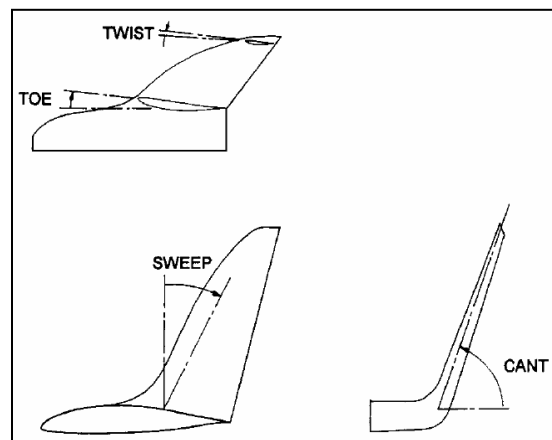


Figure 2.8 Different winglet geometry design considerations (Maughmer, 2003)

Sy et al. (2020) numerically investigated putting a split winglet (have 45° cant angle on each side) on a HAWT. In their CFD analysis, simplification was done due to limitations in computational resources. The rotor hub is hollowed out and the domain was sliced in half. The rotating and stationary domains are based on the blade length (L) of the turbine. The rotating domain has a radius of $2L$ and its depth is $2.5L$, while the stationary domain has a radius of $5L$ and depth of $9L$. Furthermore, their study also used a pressure-based steady-state RANS solver and a turbulence model of standard $k-\epsilon$. It should be noted that different turbulence models were tested, and that this turbulence model gave the closest results from experimental data. The results of the study show that putting winglets on a turbine blade reduced the swirling motion as well as the maximum tangential wind velocity up to around 20%. A single winglet tip with cant angle of 45° to the suction side showed an average of 1.23% improvement at different wind speeds, while the split winglets gave a better average improvement of 2.53%. It should also be noted that the turbines with winglets perform better at higher tip speed ratios.

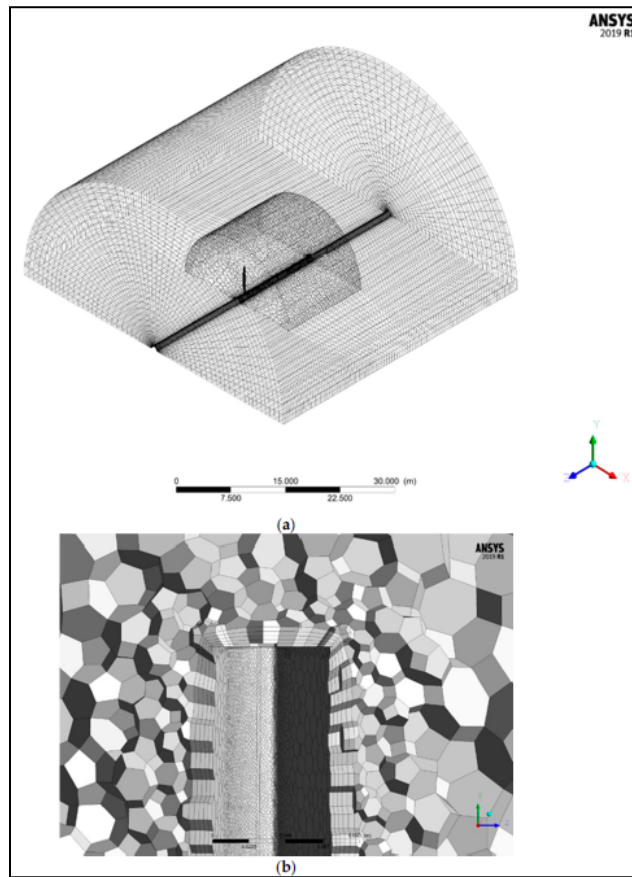


Figure 2.9 Generated mesh of (a) the entire domain and (b) near blade tip from Sy et al. (2020)

Similarly, Khalafallah et al. (2019) simulated 18 combinations of the following design parameters: winglet direction, twist angles (-2° , 2° , and 10°), and cant angles (20° , 40° , and 60°). Results show that the best performance was reached by directing the winglet on the suction side of a straight blade and using a cant angle of 60° and a twist angle of 2° , allowing a 1.75% power coefficient increase.

Khaled et al (2019) used numerical analysis to determine the optimal design for a winglet that has a design speed of 6 m/s and a rotor radius of 350 mm. This study compared blades with different winglet lengths cant angles and computational results were validated (error within $\pm 5\%$) using selected blades in a wind tunnel. Computational results show that a cant angle of 45° with a height of $0.07R$ was able to increase the produced power by 9.83% at the design TSR of 5, while a cant angle of 90° increased the power coefficient by 8.28%.

Zhu et al. (2017) also used numerical analysis (CFD) three different configurations of winglets, wherein one points up the suction side, one points down the pressure side, and the other one combines both of these in one blade tip. The flow speed is at 1.73 m/s and different hub pitch angles were also used. With all winglets at $0.025R$, 90° cant angle, and 45° sweep angle, the highest C_p increase at 15° pitch angle was by the third configuration at 3.96%, while at 20° is by the second configuration at 3.84%, and finally at 25° is by the first configuration at 2.6%. Furthermore, comparing the two single-winglet tip configurations show that when the winglet is pointing on the suction side, it is generally better with the exception of a pitch angle of 15° , while utilizing the double-tip configuration provides a general significant increase in power coefficient with the exception of TSR 7-8 at 25° pitch angle.

With the advancements in technology and manufacturing, experimental methods and validations become more prominent and accessible. Gertz D et al. (2012) on the other hand discovered that a cant angle of 90° with a length of $0.08R$, twist of -0.5° and toe angle of -0.5° was able to increase the performance coefficient (C_p) by 8%, and the maximum performance was achieved at around 7 m/s. It should be noted that this winglet also faces the suction side and that these settings are still not the optimal design due to stronger underpredicted tip vortex strength, which just means that the increase in performance coefficient could potentially have better values for better energy production and efficiency.

It should also be known that not all winglets applied on aircrafts are directly applicable due to drawbacks on turbine blades. Such a case was studied by Fathi (2020) wherein a spiroid winglet was put onto a wind turbine blade. Results of CFD indicate an increase in power output, but also shown was an increase in tip vorticity. This is important as it can affect the performance of other turbines behind it in the case of wind farms.

Notice that in these studies, wind speeds are relatively high compared to the aforementioned marine current speed of 1.73 m/s utilized by Bahaj et al. (2007) in their experimental study without winglets, and quite notably, each study had their own configuration of a winglet, showing that design considerations are on a case-to-case basis. Also, these studies also show the applicability of what Whitcomb (1976) and Maughmer (2003) suggested in aircrafts wherein tilting the winglets towards the suction side of the blades/wing increases the performance. These are just a few of the 124 studies in Scopus' November 2021 collection. This is in contrast to the huge lack of studies for winglet application for tidal turbines. Some of the parameters of the numerical analysis by Sy et al. (2020) were used in this study.

2.5 Winglet application to tidal turbines

A considerable amount of research has been conducted in order to improve the HATST energy efficiency, however there is still room to improve the energy efficiency. Engineering applications involving fluid flow around wing/hydrofoil structures have frequently encountered tip vortices. This phenomenon occurs due to the pressure difference between the pressure side and suction side of the blade. Considering the viscosity of the water, tip vortices would form behind the blade tip once the two opposite flows from both sides meet at the trailing edge and it is known to develop strong negative effect on said structures often resulting in an increase in drag, a loss of dynamic lift force, wake structure distortion, unstable operation etc. (Ren et al., 2019). Thus, many academic research studies in fluid mechanics have been conducted in order to tackle the said problem (Arndt & Maines, 1994; Çetinkaya & Oral Ünal, 2020).

An effective way to reduce the formation of tip vortices for a marine propeller is by using winglets (Hongtao et al., 2019). However, biofouling is a major concern in marine applications thus most HATST does not have winglets (*Tidal turbines: Installation & Procurement: Atlantis*

resources, 2019; Frankael, 2010; Dürr & Thomason, 2010). Since it is difficult to predict biofouling, it is much more difficult to predict how exactly it will affect the shape and performance of hydrofoils (Nedyalkov, 2015). In light of this review, the researchers would not delve into the biofouling effects of the tested design.

In the study of Hongtao et al. (2019), he made a raked tip turbine design with varying rake angles from 10° to 15° . In the computation, Shear-stress transport (SST) $k-\omega$ turbulence model was used. Results show that adding any winglet configuration significantly reduces blade tip vortices, but a higher rake angle reduces thrust coefficient. Similarly, Lee et al (2012) proposed a raked tip turbine design in order for better cavitation performance, they found that a cant angle of 90 degrees has improved C_p by 2.08% with the magnitude of vorticity reduced.

Song et al. (2012) introduced a different approach wherein the blade was just focused on being efficient, which is where NACA series airfoils are known for. It is also mentioned that because typically, wind turbine blades are composite, adopting thicker foils towards the hub to deal with fluctuating stress from unexpected gusts. A change in the hub shape was also made for the other two configurations, adopting a 2:1 ellipse for the root hub. The domain in this study was 1/3 of the full domain, adopting a 120° wedge for computational easeness. Moreover, the approach is varying TSR to calculate power coefficient. The results of the study show that utilizing the NACA-63418 throughout the blade provides improvement on efficiency, specifically 0.19% average power coefficient difference, and is more prominent on higher TSR. Putting the rake tip winglets show better efficiency due to less tip vorticity, increasing the C_p difference to 4.84%. It should also be known that the blade was also based on the study of Bahaj et al. (2007), and that the study was also experimentally validated and shown similar results, but a reduction in expected values from the simulations.

Ren et al. studied winglets in 2017 and 2019 wherein they investigated the tip chord that can increase the C_p the most at TSRs from 4 to 10. They used NACA 63-418 throughout the blade and a 2:1 ellipse on the root hub while one-third of a cylinder was used as the domain, similar to what Song et al. (2012) employed in their study.

In the 2017 study, three types of winglets were tested: trapezoid tip, triangle tip, and a hybrid winglet. The trapezoid tip is a blended winglet and bends towards the suction side, and visually leaning towards the trailing edge, at a cant angle of 90° that provided a better performance at TSRs lower than 7 compared to the base turbine, giving the best improvement of 5.03% C_p increase at TSR = 4. The hybrid winglet, resembling a split winglet, on the other hand was also a blended winglet and is swept towards the trailing edge. The one on the suction side has a 10 mm tip chord, 75° cant angle, and 22 mm winglet height, while the split on the pressure side has an 8 mm tip chord, 85° cant angle, and 16 mm winglet height. This configuration gave the same trend of increasing the power coefficient at TSRs lower than 7, but the trapezoid tip performed better at those TSRs. Both configurations did not have a positive effect on TSRs above 7, with the hybrid tip being more forgiving compared to the trapezoidal tip. The best configuration was the triangle tip, which is also bent towards the suction side at an angle of 90° . Instead of a blended winglet, it has a cross-section of an 18:1 ellipse and has a 2 mm tip chord with a winglet height of 22 mm. This winglet configuration provided an increase in power coefficient at all TSRs from 4 to 10, giving the best increase of 14.11% at TSR 10, and at the optimum TSR of 5 by 4.34% C_p increase. Furthermore, all winglet configurations reduced the tip vortices leading to an increase in efficiency.

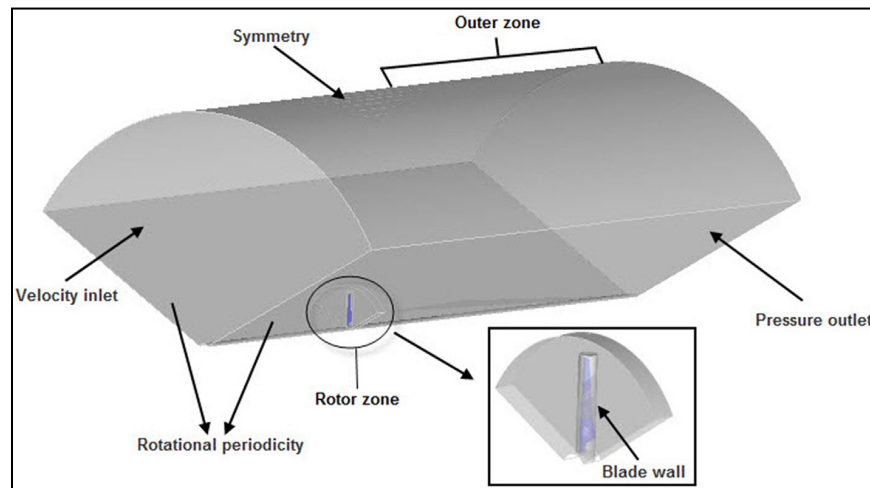


Figure 2.10: Computational domain with boundary conditions (Ren et al., 2017)

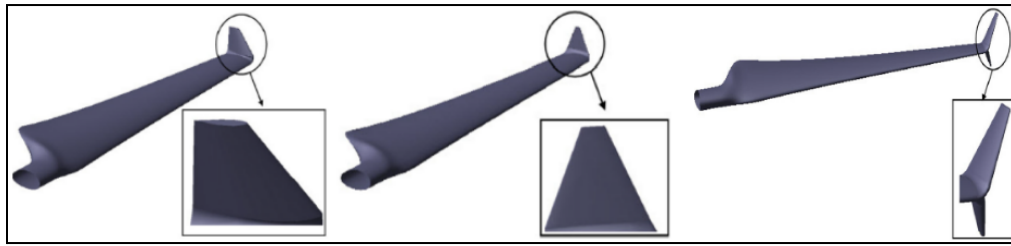


Figure 2.11 Three types of winglets: trapezoidal tip (left), triangular tip (middle), split tip (right) from Ren et al. (2017)

The 2019 study by Ren and his colleagues did a similar study but this time employed a 16:1 elliptical profile for the blended winglet configurations investigated, which was proven to be better from their 2017 study than blended winglets. These winglets have a cant angle of about 90° , tip chord of 8 mm and a winglet height of 15 mm. Their results show a better increase when the said winglet was bent towards the suction side, providing a 4.34% increase at the optimum TSR of 5. Moreover, upon doing configurations on the tip chord, this value further increased to 4.66% when the winglet had a tip chord of 16 mm at the optimum TSR but was the least effective compared to shorter tip chords at TSRs above 5.4. It was also determined that the tip chord should not be above 8 mm to increase the power production throughout all TSRs. They also did configurations on the winglet height and determined that to get a positive increase in all TSRs included, the winglet height should not be less than 17 mm. Another configuration was also checked wherein different cant angles were investigated, and results show that the cant angle of a winglet should not be less than 80° in order to get a positive increase for all TSRs examined.

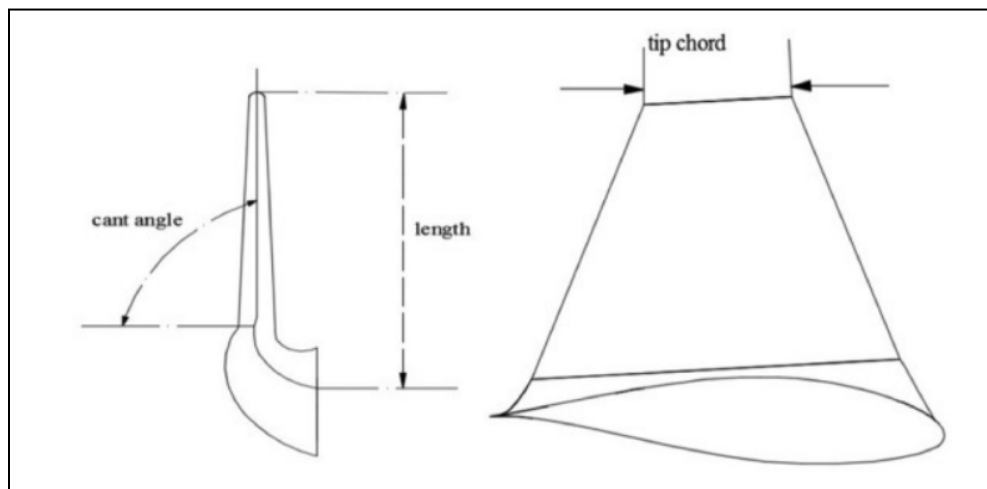


Figure 2.12 Main parameters for blended winglet design. Image Source: Ren et al. (2019)

In all of the mentioned studies, all of them utilized similar boundary conditions for CFD: inlet as uniform velocity, outlet as pressure outlet, and mesh between the stationary and rotating domain are set to be interface. Moreover, due to most HATSTs not having winglets, there is a limited number of studies regarding this topic. To add onto this, this study intends to investigate the effect of winglets on the hydrodynamic performance of a tidal turbine using numerical methods, particularly computational fluid dynamics (CFD).

2.6 Integration

The study of adding winglets on tidal turbine blades has only few directly related references. But even so, many of the mentioned references were used as guides and integrated into the methodology of the study. Firstly, the main profile of the blade that was employed in this study is from the experimental investigation of a tidal turbine by Bahaj et al. (2007) as well as the environmental conditions of the cavitation tunnel, particularly the flow speed. This is because the blade from Bahaj et al. (2007) employs NACA airfoils, and was also used in the study of Song et al. (2012) which utilized an elliptical root hub on their configuration's second case. The mesh made by Ren et al. (2017) and Sy et al. (2020) were the basis for the computational domain for CFD analysis, while the hollowing of the rotor hub was made similar to the latter study. The winglets and their configurations that were employed in this study are based on the two studies by Ren et al. in 2017 and 2019.

III. METHODOLOGY

The hydrodynamic performance of a HATST is investigated via CFD in this study. In order to validate the CFD model, comparisons with experimental data are made. Upon validation, winglets are added to the base design and their effects to the performance of the turbine are investigated.

3.1 Turbine Blade Selection

The blade used in this study is based on the blade used by Bahaj et al. (2007). The tidal turbine has a 3-blade rotor with a diameter of 800mm. The blade was developed to have the NACA 63-8xx profile and a chord, thickness and pitch distribution as shown in Table 2. The blade profile will have 2D section coordinates interpolated from coordinate-based data for NACA 63-812, 63-815, 63-818, 63-821, and 63-824 for 17 stations along the blade.

Table 2.1 Blade geometry specifications (Bahaj et al. 2007)

r/R	Radius (mm)	c/R	Pitch distribution (deg)	t/c (%)
0.2	80	0.125	15	24
0.25	100	0.1203	12.1	22.5
0.3	120	0.1156	9.5	20.7
0.35	140	0.1109	7.6	19.5
0.4	160	0.1063	6.1	18.7
0.45	180	0.1016	4.9	18.1
0.5	200	0.0969	3.9	17.6
0.55	220	0.0922	3.1	17.1
0.6	240	0.0875	2.4	16.6
0.65	260	0.0828	1.9	16.1
0.7	280	0.0781	1.5	15.6

0.75	300	0.0734	1.2	15.1
0.8	320	0.0688	0.9	14.6
0.85	340	0.0641	0.6	14.1
0.9	360	0.0594	0.4	13.6
0.95	380	0.0547	0.2	13.1
1.0	400	0.05	0	12.6

3.2 Turbine Blade Modeling

The airfoil coordinates for the blade were obtained with the help of JavaFoil and QBlade. These coordinates were then transferred to Fusion360 where the blade's body was created using the loft command. A 2:1 ellipse is the hub fitting section shape similar to that employed by Ren et al. (2019). The blade is shown in Figure 3.1.

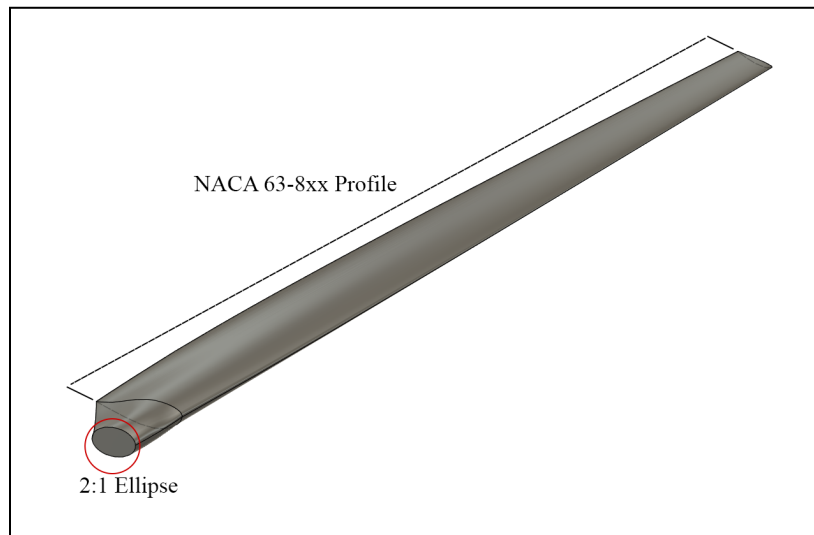


Figure 3.1 Tidal turbine blade

3.3 Numerical Model

The blade models were then used as reference to create the rotating domain, which was further enclosed by a stationary domain using Fusion 360. The rotating domain has a radius of 2 times and a depth of 2.5 times the blade length, while the stationary domain has a radius of 5

times and a depth 9 times the blade length. In order to simplify the CFD analysis and save computational resources, a third of the cylindrical domain was used. The core where the rotor hub is located was then hollowed out to create a 120° wedge model, using periodic boundary conditions on both ends of the domain.

The stationary domain contains the velocity inlet, pressure outlet, and outer wall. The inner rotating domain contains the blade. The stationary domain was meshed using Ansys Mechanical Meshing, while the rotor domain was meshed using Ansys Fluent Meshing. The model was meshed using the curvature size function. Minimum size was set to 0.4 mm while the curvature normal angle was set to 1 degree. Growth rate was set to 1.1. An inflation layer was also used. It contains 10 layers with a growth rate of 1.4 and a first layer thickness of 0.01 mm. The first layer thickness was calculated to have a Yplus (y^+) of less than 5. The mesh for the rotor domain was immediately converted to polyhedra mesh to reduce cell count. The resulting mesh with boundary conditions is shown on Figure 3.2.

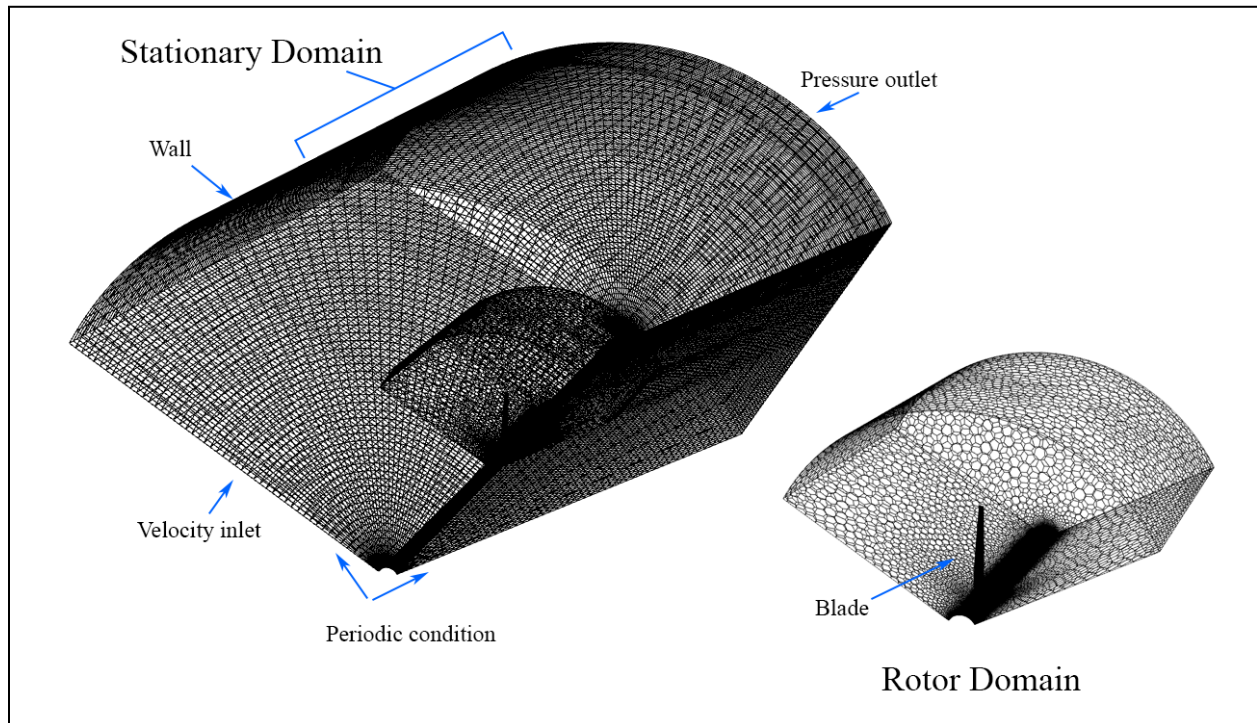


Figure 3.2 Mesh generated for the computational domain with boundary conditions

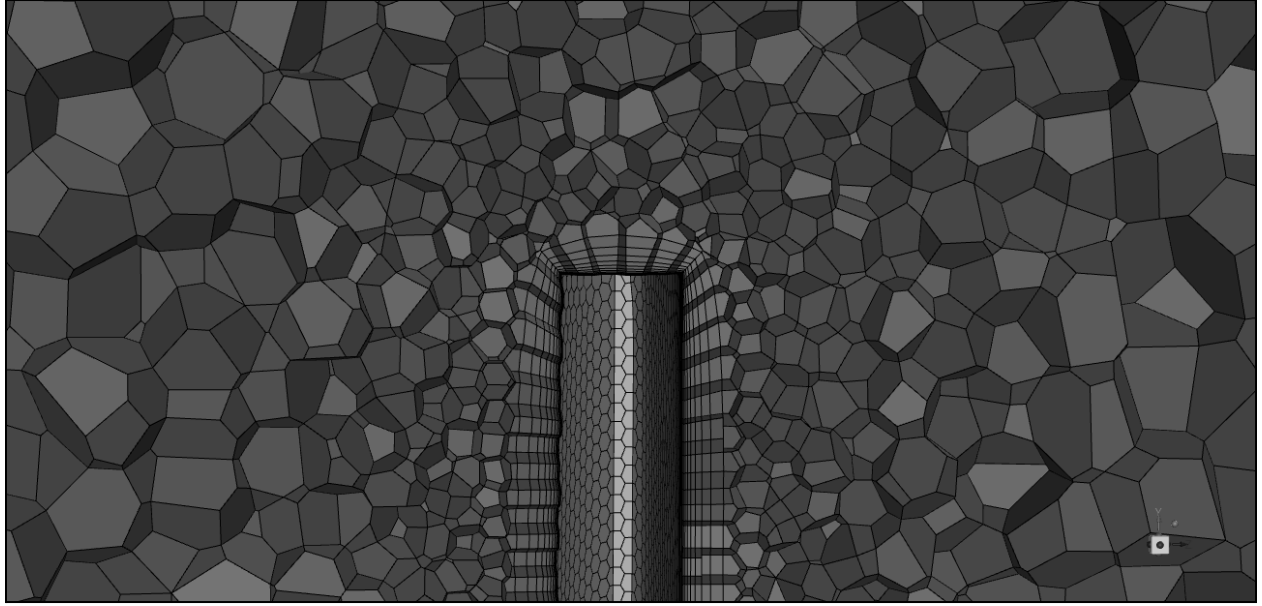


Figure 3.3 Close-up of the mesh near the blade tip

3.4 Solution and Validation

The converted mesh has 2.44 million cells, 11.54 million faces, and 7.24 million nodes with a minimum orthogonal quality of 0.326 and a maximum aspect ratio of 39.98. The simulation uses a pressure-based Reynolds-averaged Navier-Stokes (RANS) Solver. Both the stationary and rotor domain were set as fluid domains with water as the material ($\rho = 998.2 \text{ kg/m}^3$). Turbulence model was set to k-omega SST. Boundary conditions include non-slip walls for the blade, hub, and outer wall, periodic for the side walls, inlet velocity of 1.73 m/s for the tunnel entrance, and pressure outlet for the tunnel exit. Mesh interface was also set in between the two domains. In the solution section, the coupling scheme was set to “coupled” while discretizations were all set to second order to improve accuracy of the solution. The transient simulation was set to run for four rotations with a timestep size of 1° per timestep. A report definition to monitor the coefficient of moment on the rotor blade along the axis of rotation was created. Simulations are deemed to be converged when residual values fall below 10^{-5} .

The power coefficient (C_p), a non-dimensional number that represents the fraction of power extracted by the rotor from the water, can be computed using Equation (1), where (τ) is the torque, (ω) is the rotor’s angular velocity, (ρ) is the water density, (A) is the area of the rotor

disk, and (V) is the water velocity. Torque can be computed from the coefficient of moment (C_m) values obtained from the simulation using Equation (2).

$$C_P = \frac{\tau\omega}{\frac{1}{2}\rho AV^3} \quad (1)$$

$$\tau = \frac{1}{2}\rho C_m V^2 \quad (2)$$

The C_p values obtained from the base blade simulation are then compared to the experimental values for validation as shown in Figure 3.4. C_p values from TSR 6-10 show a percent error of up to 6.9 % while TSR 4 showed a 25.6 % difference. This model was deemed acceptable for the current study because only one of the points deviated relatively from the rest.

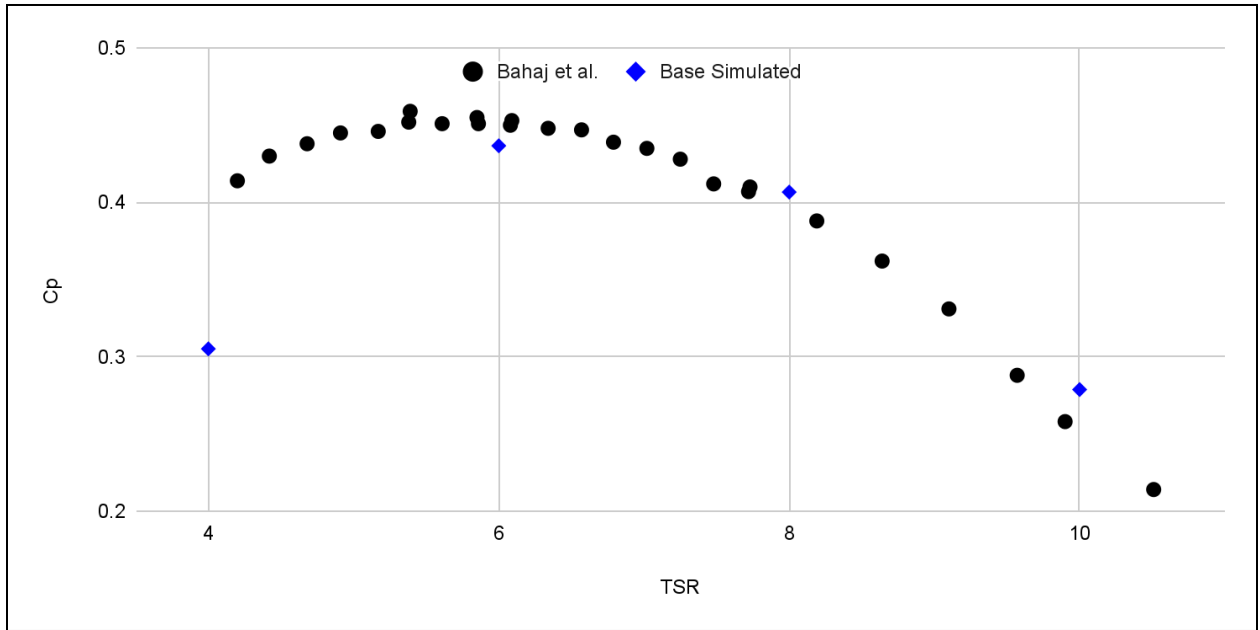


Figure 3.4 Validation of the base simulation vs. experimental results by Bahaj et al. (2007)

3.5 Addition of Winglets

Upon verification of the simulation model, winglets can now be added to investigate its effects via CFD. Since adding a winglet also results in an increase in redundant area, and the profile drag would be increased, a design concept of the chosen winglet should ensure that the

profile drag of the winglet is smaller than the decrease in the induced drag (Maughmer, 2003). Thus the desired design chosen is an application of the winglet tip design that is based on the blended winglet tip design considerations of Ren et al. (2019) will be implemented as shown on Figure 3.5 and 3.6.

Respectively, two design considerations were utilized by the researchers. The trapezoidal design consideration has a tip chord of 8 mm and a length of 15 mm, with a consistent cant angle of 90° bent towards the suction side. The section shape of the winglet took the profile of the ellipse with 16:1 major to minor axis ratio. The centerline of the winglet is tangent to the center of the chord of the blade tip.

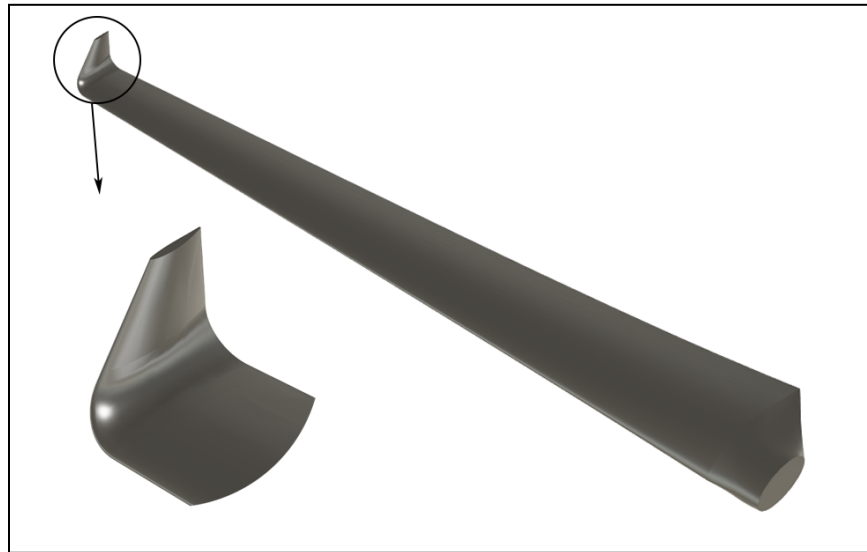


Figure 3.5 Trapezoidal winglet configuration based from Ren et al. (2019)

The other design consideration would be a triangle tip turbine without the sharp corner. Similar to the previous design the cant angle will be kept at approximately 90° bent towards the suction side and the centerline to be tangent to the center of the original tip. The section shape of the winglet will use an 18:1 ellipse with the tip chord of the winglet at 2 mm and a length of 22 mm.

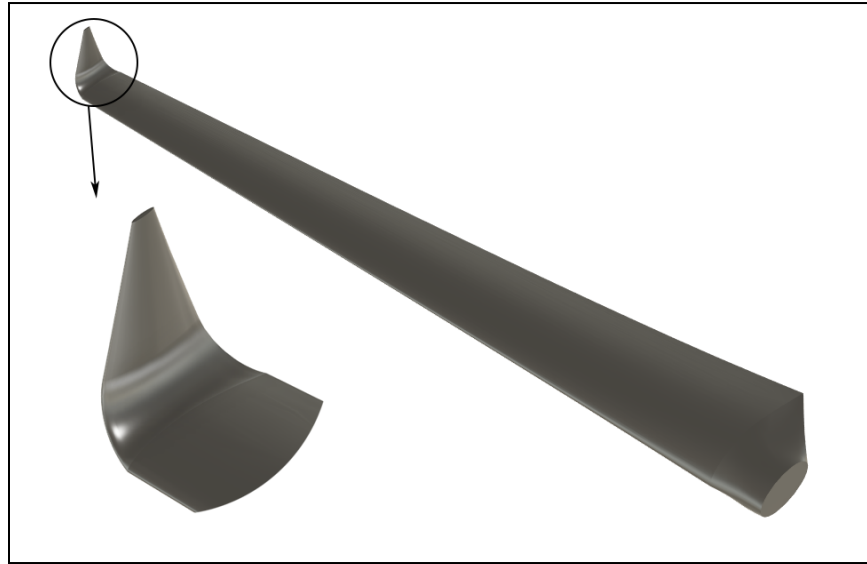


Figure 3.6 Triangular winglet configuration based from Ren et al. (2017)

In both designs, the intersection between the blade tip and the winglet is a smooth curve (employ blended winglet) instead of a sharp angle to reduce the intersection drag at the junction of the blade and winglet (Ren et al., 2019). This is made possible by inserting the said parameters in Fusion 360 and then sweeping to create the desired winglet whilst ensuring that the blade length remains constant.

IV. RESULTS AND DISCUSSION

The following are the results from the simulation using Ansys Fluent of the three types of rotor blades modeled: base model, trapezoidal winglet, and triangular winglet. Each rotor blade was run on four different tip speed ratios, 4, 6, 8, and 10, totalling to 12 data points.

4.1 Coefficient of Moment Convergence

The main data obtained from the simulations needed to compute for the coefficient of power is the coefficient of moment (C_m). This value is monitored throughout the simulation and is deemed converged when the C_m values flatten out at the end of the simulation as shown in Figure 4.1. The C_m convergence of the other simulations show similar behavior and is shown in figures A.1, A.2, and A.3 of the appendices.

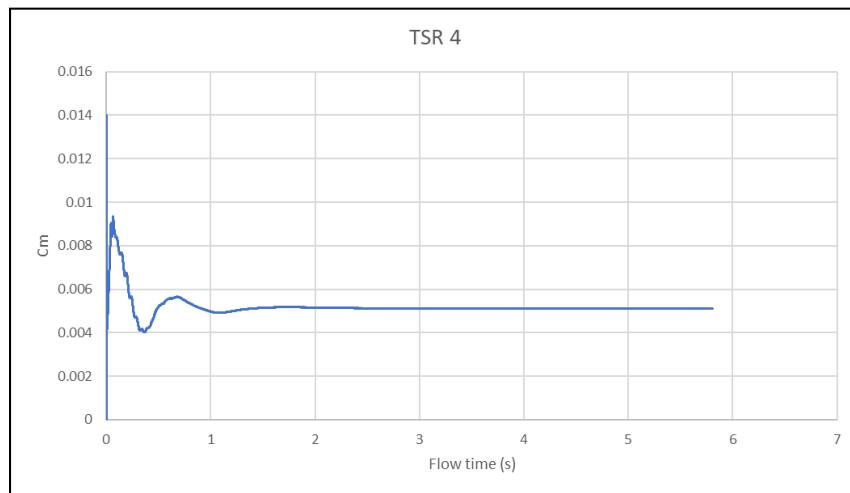


Figure 4.1 Coefficient of Moment Convergence vs flow time at TSR 4 for the base configuration

4.2 Wall Yplus

As mentioned previously, the mesh has a first layer thickness of 0.01 mm, which corresponds to a computed Yplus (y^+) value of less than five. Figure 4.2 shows the y^+ values on the blade for the base configuration. It shows that the simulation is able to achieve y^+ values of less than five for most TSRs with only a few points on TSR 10 exceeding five. This behavior is also exhibited on the trapezoidal and triangular winglet blades as shown in figures B.2 and B.3 of the appendices.

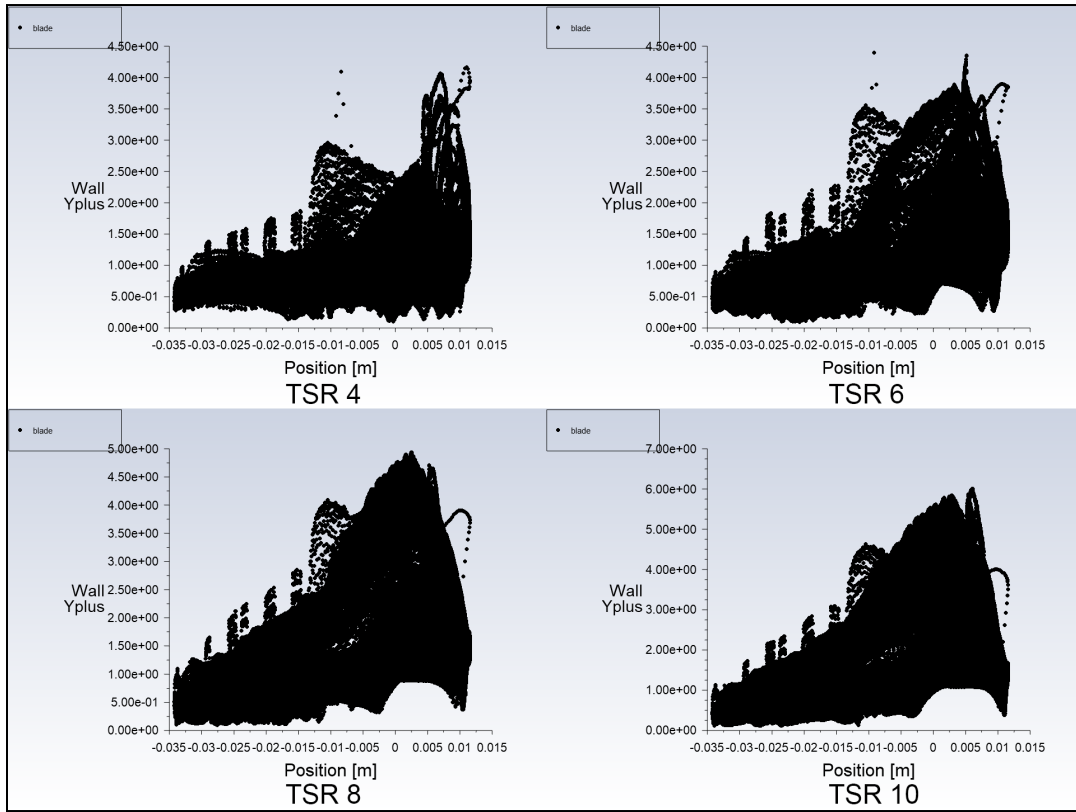


Figure 4.2 Yplus values at various Tip Speed Ratios for the base configuration

4.3 Power Coefficient

Comparisons of the power coefficient (C_p) between experimental, base, and winglet designs are shown on Figure 4.3. The simulated data all show similar behavior versus TSR. They all acquired peak power coefficient at approximately $TSR = 6$. Both winglet designs demonstrate an increase in power coefficient with an average increase of 3.71% for the trapezoidal configuration and 2.86% for the triangular configuration. Both winglet designs also have a higher increase in power coefficient on higher TSRs.

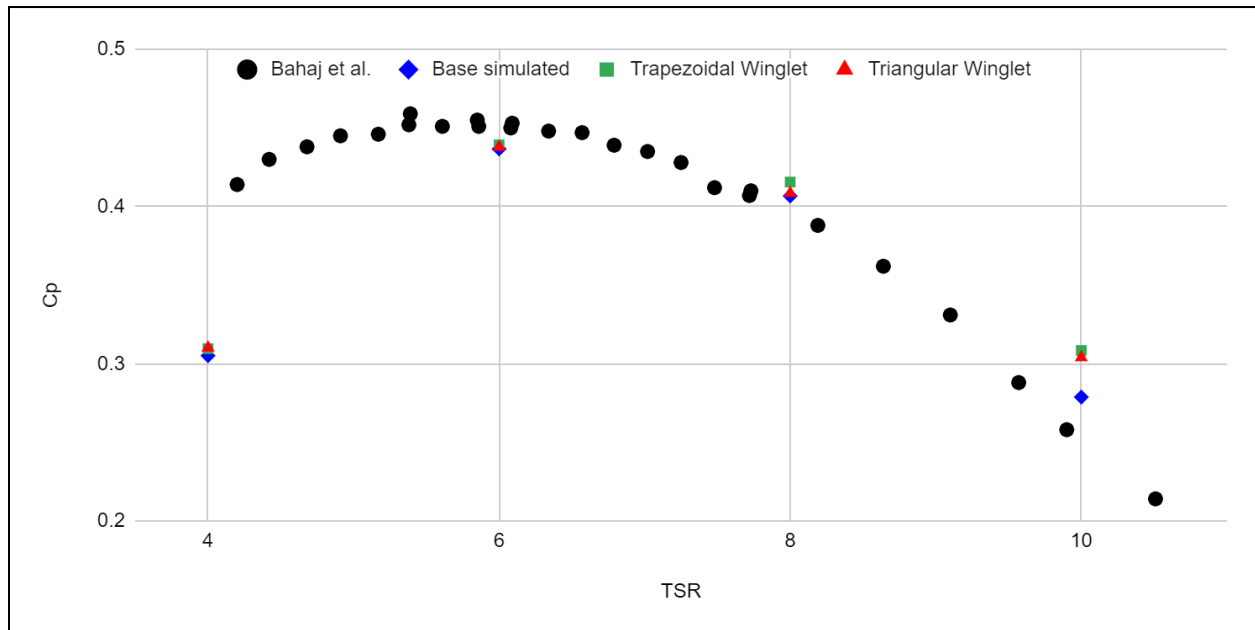


Figure 4.3 Power Coefficient comparison between experimental and simulated runs

4.4 Tip Vorticity

To observe the effects of winglets to vorticity, snapshots of the vortex core region using the swirling strength detection method with a level of 0.001 is shown in figures 4.4 and 4.5. Winglets are shown to reduce the vorticity with the triangular winglet showing the strongest reduction. Higher TSRs also show relatively smaller vorticity. The low-pressure region in the suction side of both winglet designs also shows greater magnitude compared to the base blade, this suggests higher lift generation.

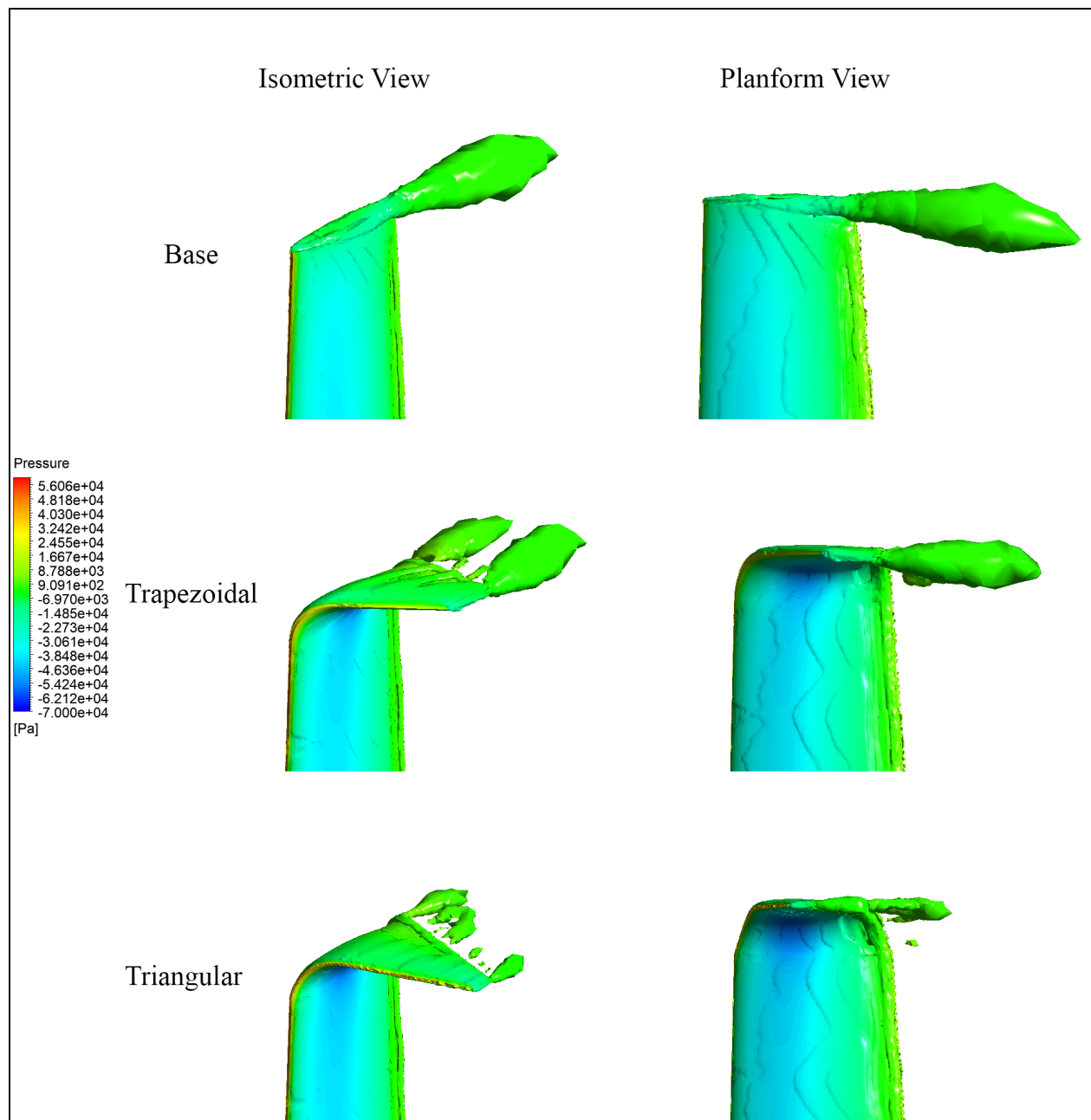


Figure 4.4 Vortices at the rotor blade tips at TSR 6

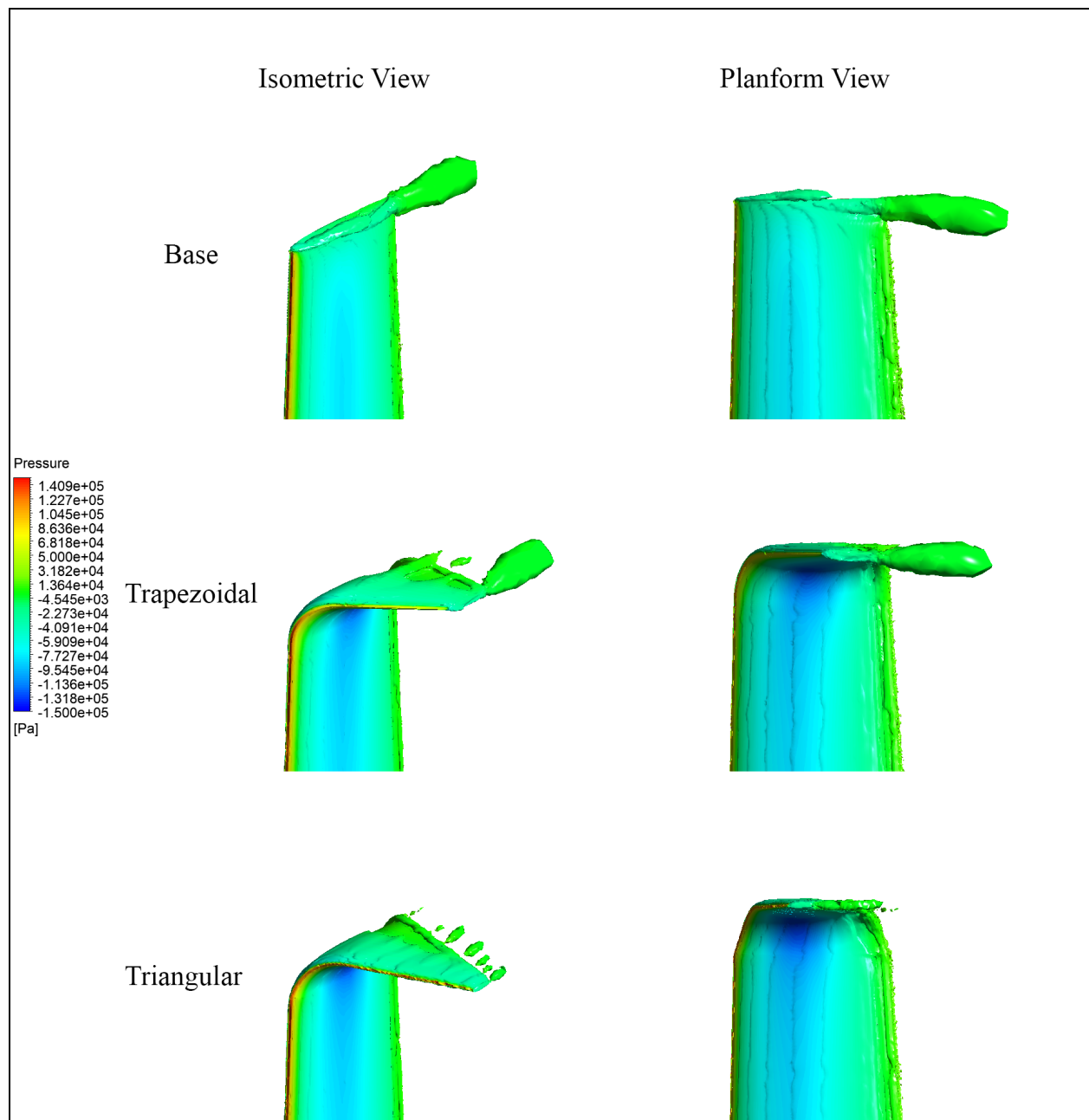


Figure 4.5 Vortices at the rotor blade tips at TSR 10

4.5 Surface Pressure Coefficient

To better see the effect of winglets on the base blade, the pressure coefficient at different locations were checked. Comparing these distributions, changes are more observed at locations nearer the blade tip ($0.95r$) and nearer the blade root ($0.25r$). This is better shown at higher TSRs especially at TSR 10 (Figure 4.7) where winglet changes are much more evident compared to lower ones such as on TSR 6 (Figure 4.6) wherein the distributions are almost similar on every configuration. Continuing the observation on the distribution at TSR 10, the addition of winglets increase the pressure coefficient at the pressure side of the blade, while also decreasing the pressure coefficient at the suction side of the blade. This increases the gap, and consequently the curve area of the pressure coefficient, thus generating a higher torque near the blade tip region and a better performance (Ren, 2019).

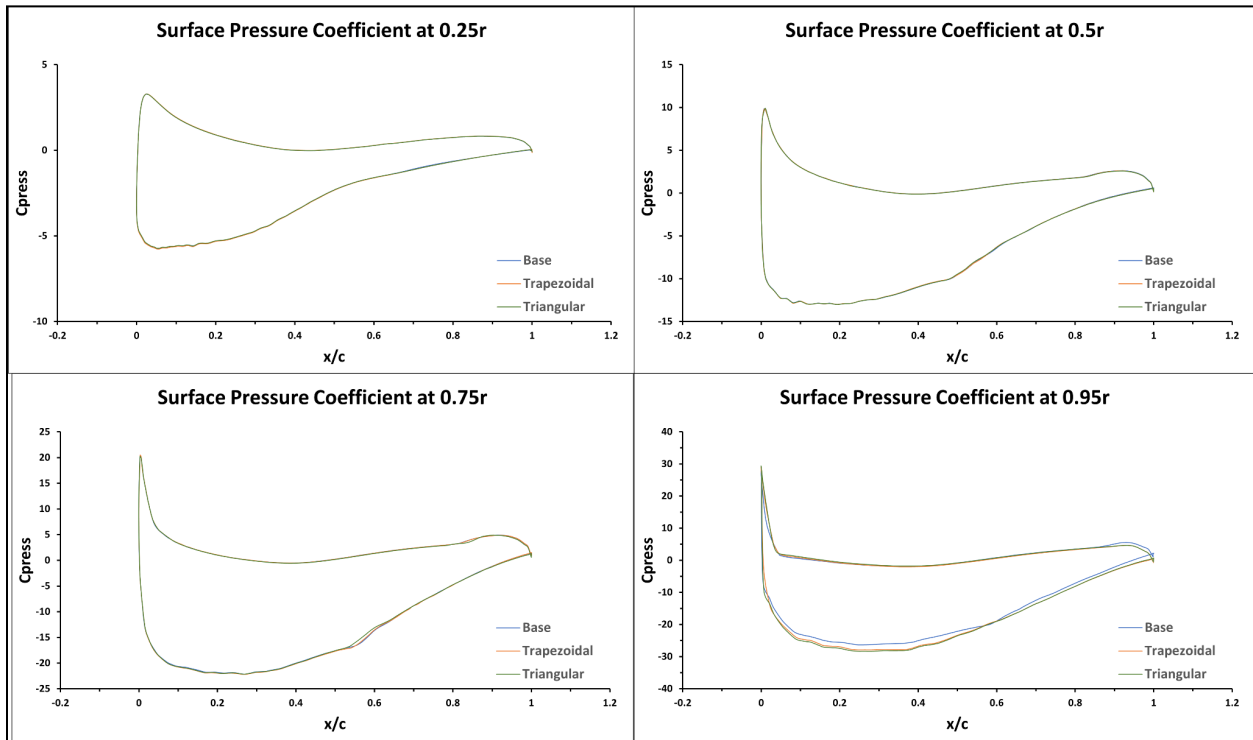


Figure 4.6 Surface Pressure Coefficient Distribution at various blade locations at TSR 6

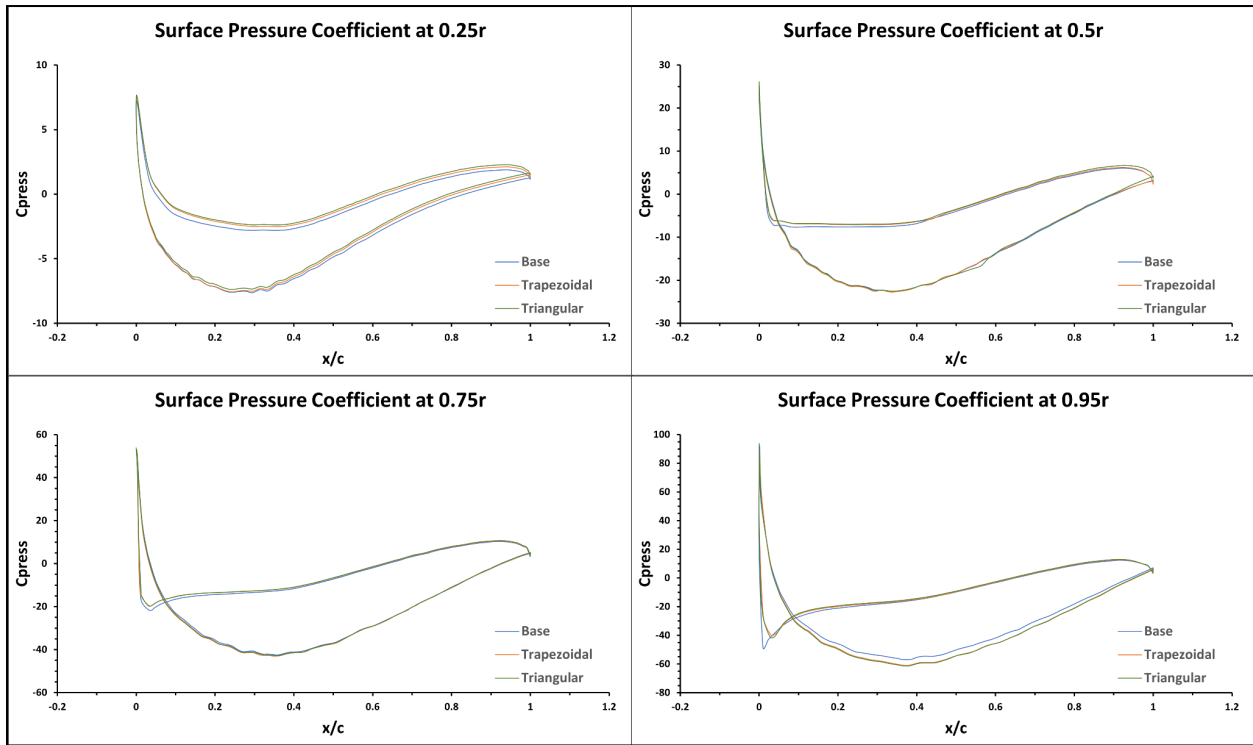


Figure 4.7 Surface Pressure Coefficient Distribution at various blade locations at TSR 10

In general, both winglet configurations contributed positively to the performance of the blade, but between the two winglet configurations, the triangular winglet performed better than the trapezoidal winglet. Additionally, all blade configurations followed the same pattern, where towards the trailing edge, the difference between the pressure coefficients decreases. It can also be inferred from the observations above that the two winglet configurations are more effective at high TSR, thus giving a more observable distribution difference.

Moreover, as expected from the results of the surface pressure coefficient, there is a little difference between the pressures when looking at the pressure contours on Figures 4.8 and 4.9. It can be seen that there is a decrease in pressure experienced on the suction side from the base blade when winglets are applied on it, and is much clearer on higher TSRs (e.g. TSR 10).

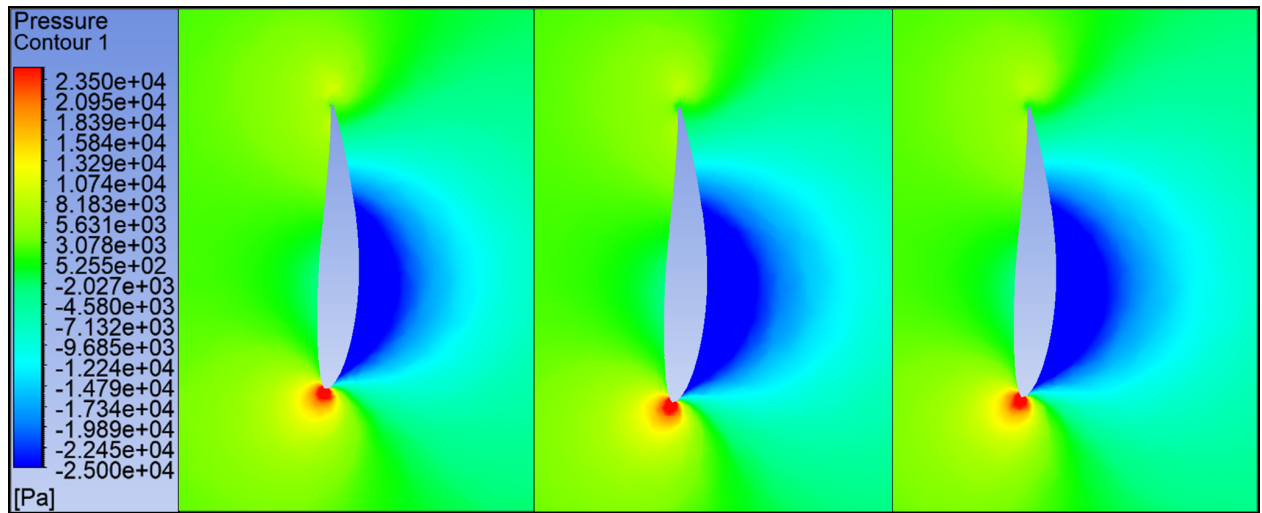


Figure 4.8 Pressure Contour at 0.95r and at TSR 6 (Base/Trapezoidal/Triangular)

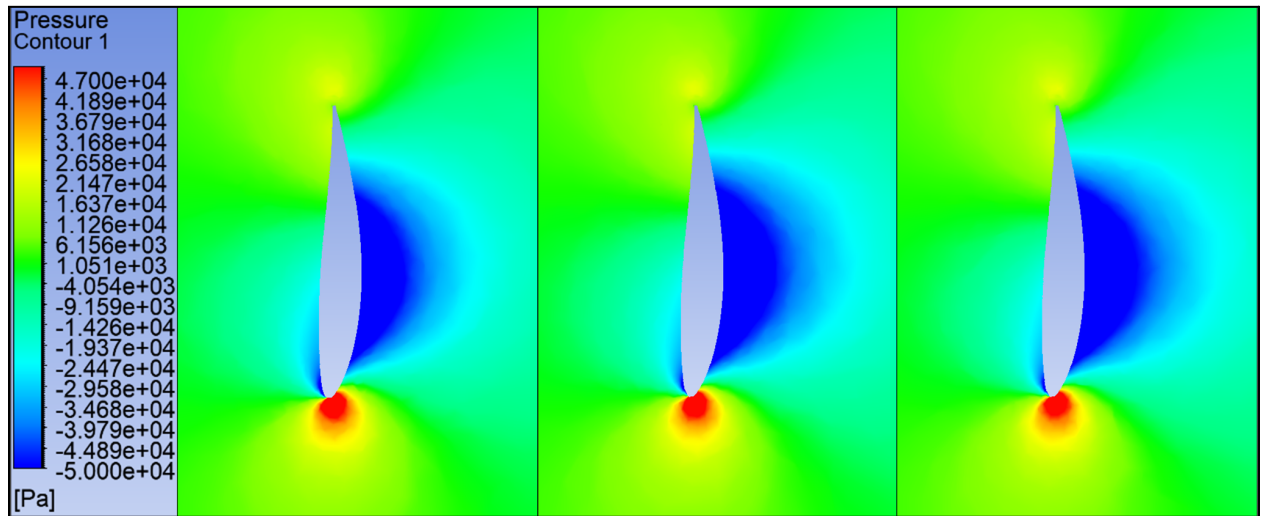


Figure 4.9 Pressure Contour at 0.95r and at TSR 10 (Base/Trapezoidal/Triangular)

V. CONCLUSION

The influences of winglets on the hydrodynamic performance of HATST are investigated in this study. Two conceptual designs with the winglet bent towards the suction side are implemented for the turbine. Results show that both winglet designs improve the performance of the turbine, giving an average of 3.71 % increase in power coefficient for the trapezoidal winglet design and an average of 2.86 % for the triangular winglet design. Both designs also show a higher increase in C_p on higher TSRs. The increase in performance was achieved via the reduction of vortices on the winglet designs and is further evidenced by the improvement of the pressure difference between the pressure and suction sides of the blade. This increased the generated torque especially near the blade tip, and is more apparent on higher TSRs.

5.1 Recommendations

The study was limited to the winglets' effects on the blade of Bahaj's study. Optimizing the winglet configurations such as the appropriate cant angle, winglet height, and the likes might provide a more accurate understanding of the winglets' effects on Bahaj's blade. Furthermore, investigation on higher TSRs could also provide further insights with regards to the effectiveness and confirm the applicability of these winglets on specific or limited tip speed ratios. As this study was done with limited computing power, it is recommended to add more parameters, i.e. resulting thrust and drag, and an investigation on better turbulence models, to provide better and more accurate results and observation.

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APPENDIX A - Convergence Graphs

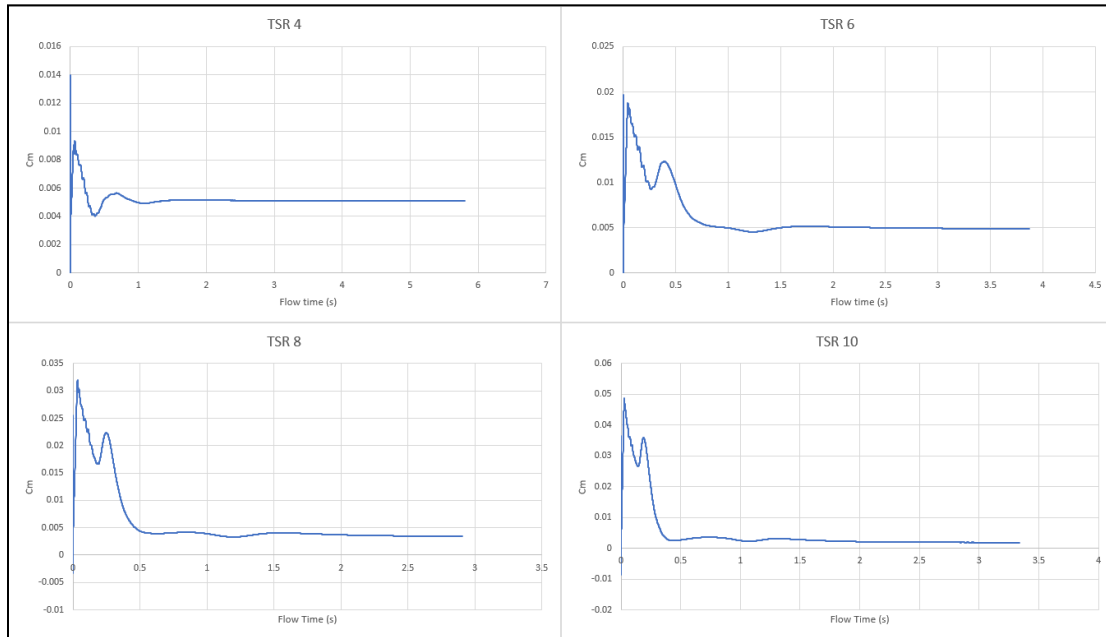


Figure A.1 Coefficient of Moment convergence over flow time at various Tip Speed Ratios for the base configuration.

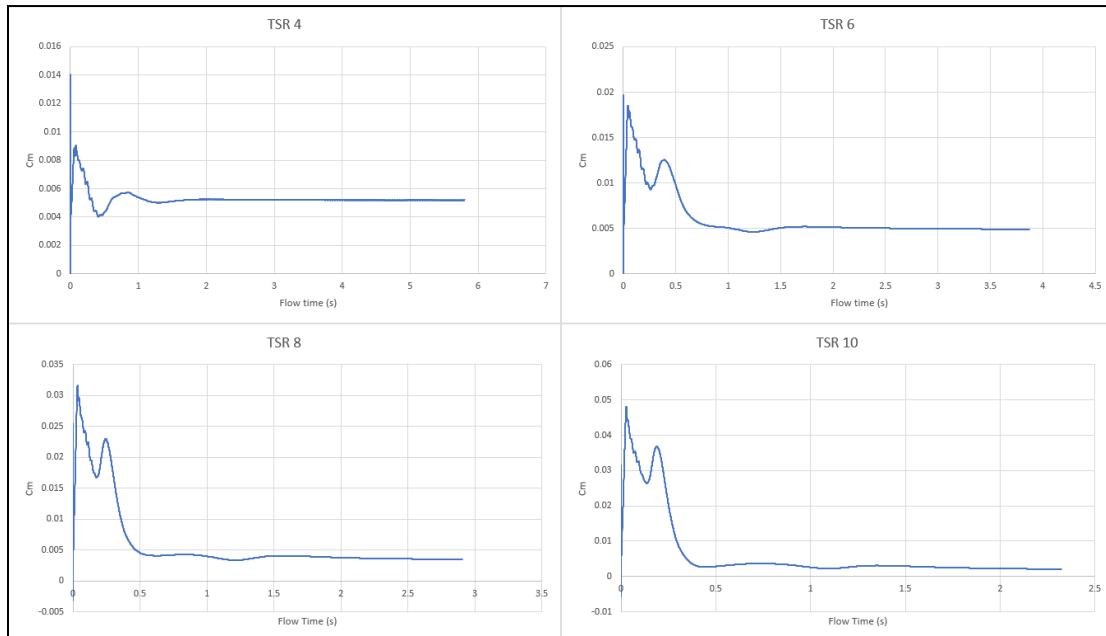


Figure A.2 Coefficient of Moment convergence over flow time at various Tip Speed Ratios for the trapezoidal configuration.

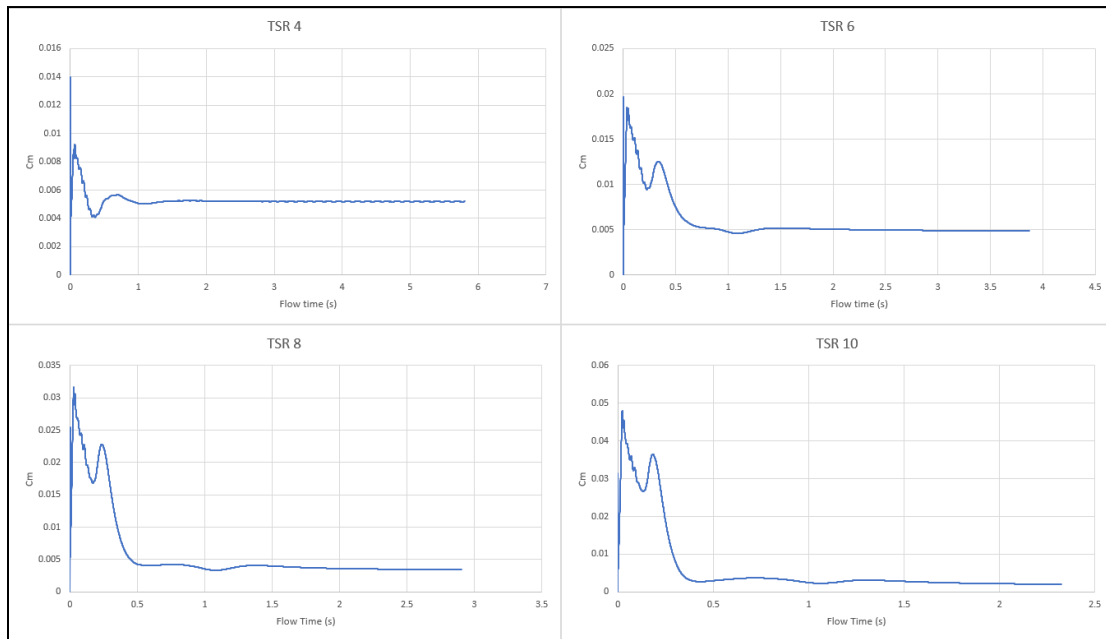


Figure A.3 Coefficient of Moment convergence over flow time at various Tip Speed Ratios for the triangular configuration.

APPENDIX B - Wall Yplus Distribution

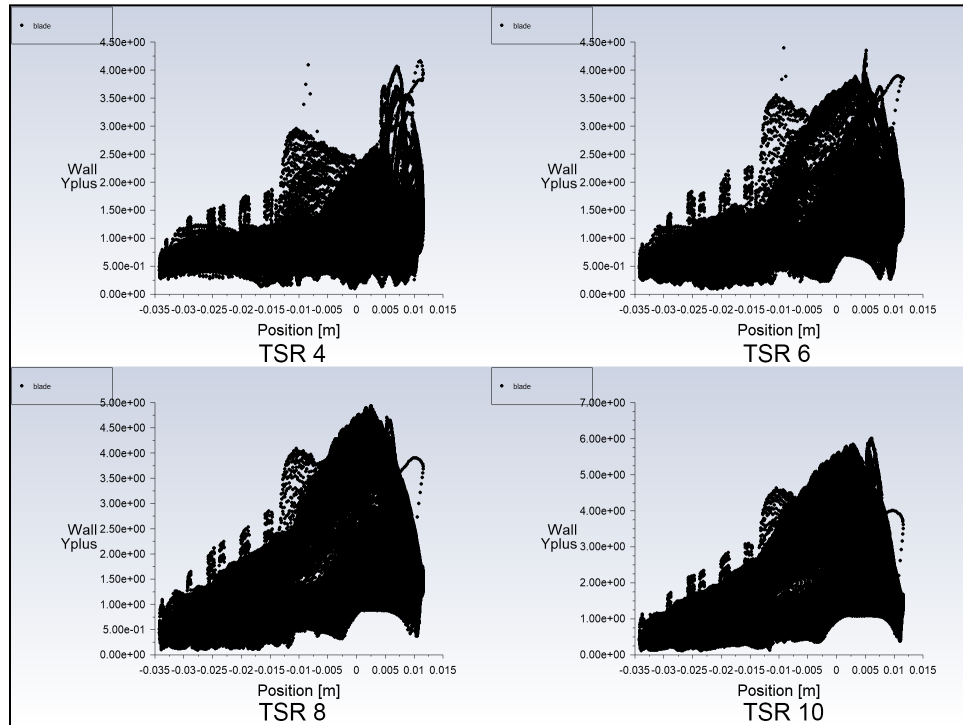


Figure B.1 Yplus values at various Tip Speed Ratios for the base configuration

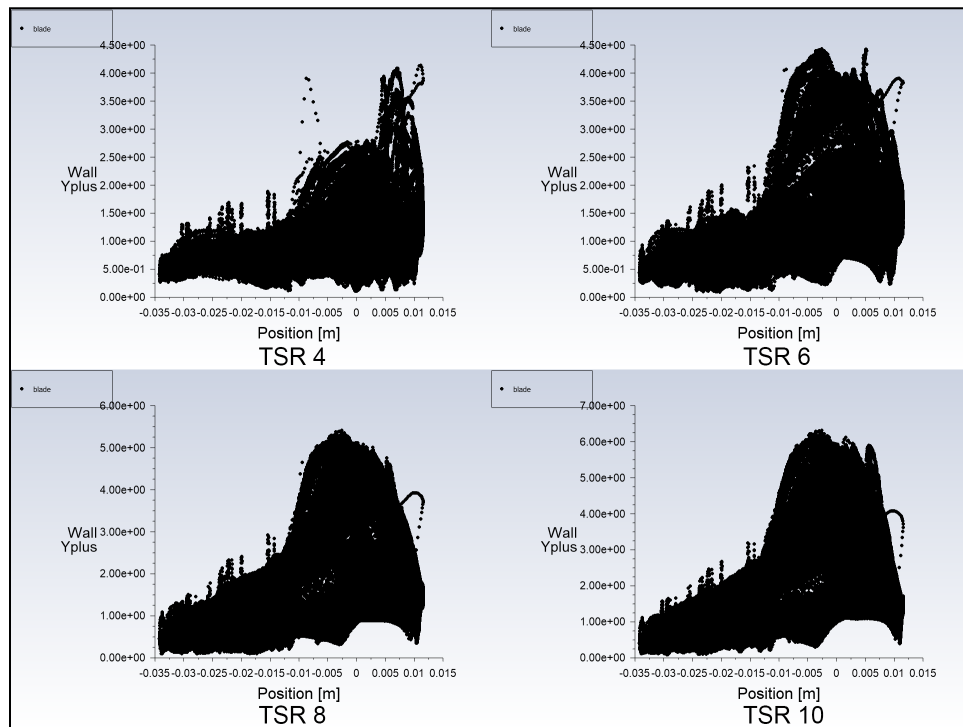


Figure B.2 Yplus values at various Tip Speed Ratios for the trapezoidal configuration

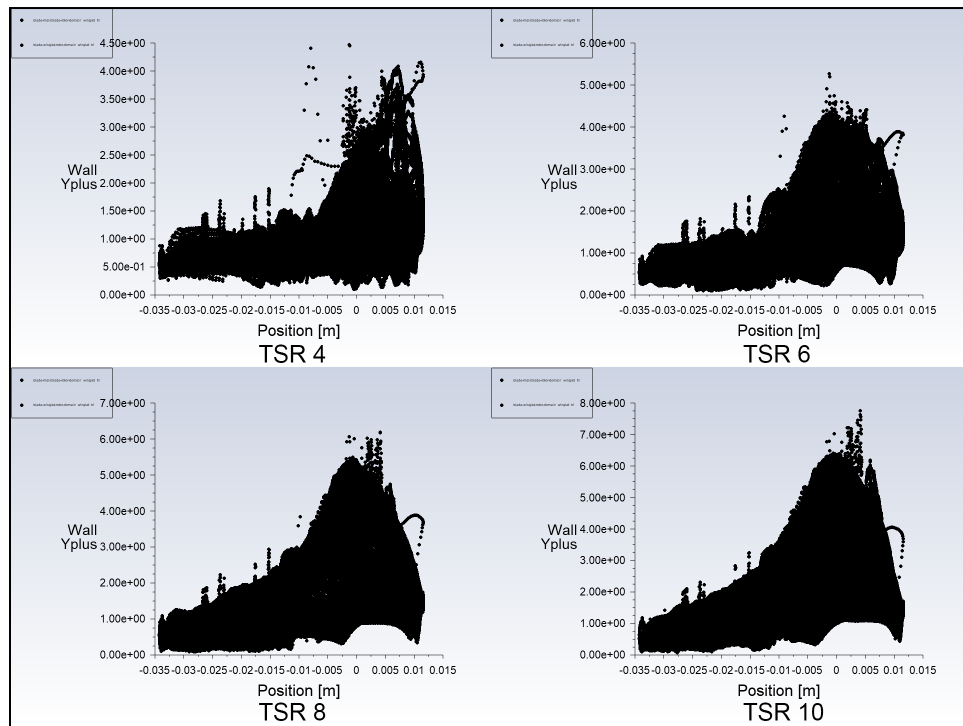


Figure B.3 Yplus values at various Tip Speed Ratios for the triangular configuration

APPENDIX C - Surface Pressure Coefficient Graphs

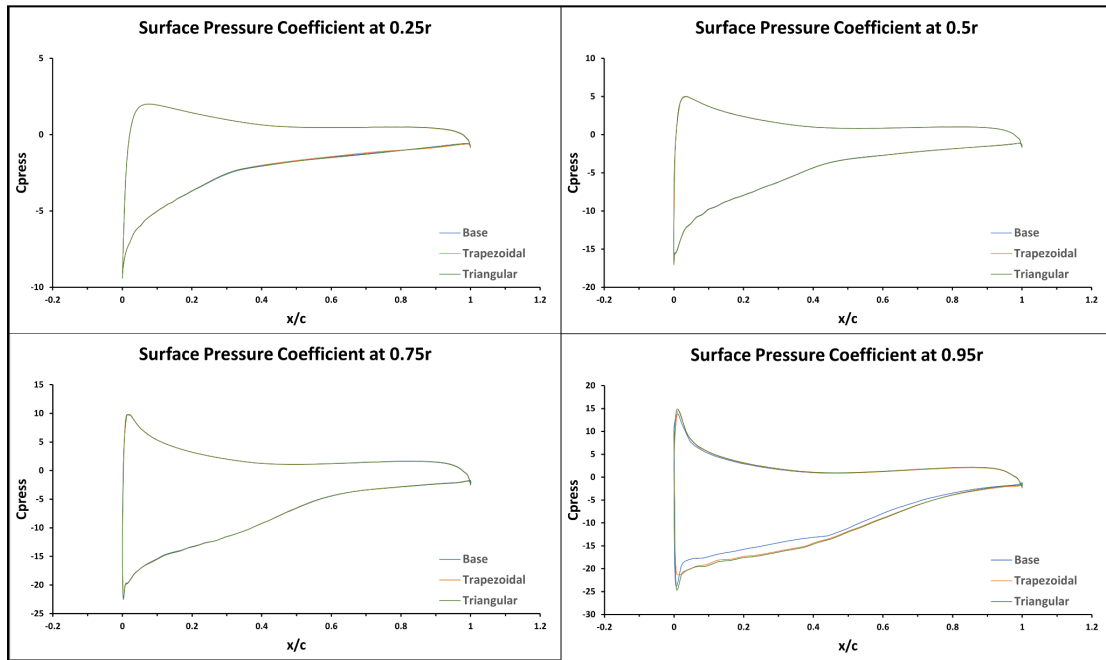


Figure C.1 Surface Pressure Coefficient Distribution for TSR 4

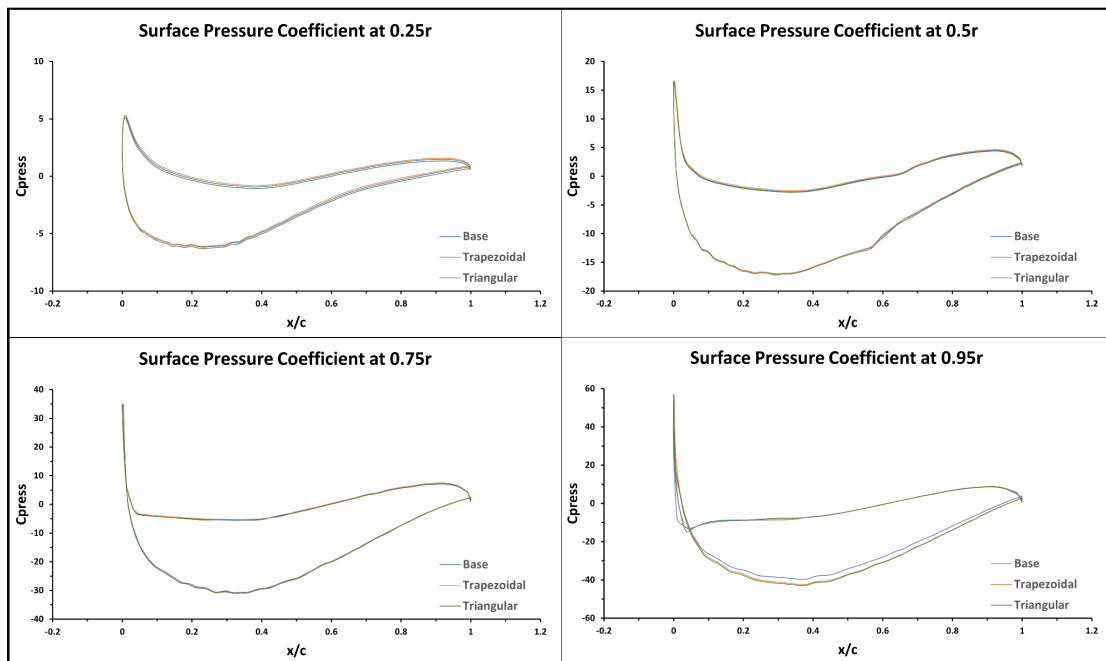


Figure C.2 Surface Pressure Coefficient Distribution for TSR 8